# Monetary Policy and Income Inequality: The Role of Corporate Earnings Allocation

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**Abstract**

This paper examines how firms' responses to monetary policy shape its distributional effects through wage and dividend payments. Using administrative tax records from Taiwan for 2004–2023 that link firms to the workers and shareholders they pay with high-frequency monetary policy shocks, we find that expansionary monetary policy raises firm revenue and increases payments to both workers and shareholders, but dividends respond far more strongly than wages. The dividend response is stronger among firms with greater reliance on short-term borrowing, consistent with a credit channel in which monetary easing relaxes financing constraints and allows firms to raise shareholder payouts. These firm-level responses translate into unequal income gains. The total income gain of the top 1% is about 2.2 times that of the bottom decile, and dividends account for about 61% of the top 1%'s gain. Monetary easing in Taiwan thus disproportionately benefits top earners, primarily through dividend income.

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# 1 Introduction

A growing body of empirical research documents that monetary policy generates heterogeneous effects across households, with wage and wealth gains distributed unevenly across the earnings and wealth distributions (Coibion et al., 2017; Mumtaz and Theophilopoulou, 2017; Holm, Paul and Tischbirek, 2021; Amberg et al., 2022; Andersen et al., 2023). Theoretical frameworks attribute these differential effects to heterogeneity in household financial constraints, asset holdings, and labour market attachment (Kaplan, Moll and Violante, 2018; McKay and Wolf, 2023; Milani, 2026). An important gap in this literature, however, is that it focuses almost exclusively on individual-level outcomes and abstracts from the role of firms, which are the direct source of both labour and capital income. This omission is nontrivial: when monetary easing raises firm revenue or lowers financing costs, how firms divide these gains between wages and dividends determines who ultimately benefits. Existing evidence shows that firms share only a limited fraction of profit gains with their workers (Card et al., 2018; Kline et al., 2019), raising the possibility that the gains from monetary easing accrue unevenly to workers and shareholders. This firm channel may be further shaped by differences across firms. Firms respond heterogeneously to monetary policy depending on their financial constraints, leverage, and investment levels (Bernanke, Gertler and Gilchrist, 1999; Ottonello and Winberry, 2020; Gnewuch and Zhang, 2025), which shapes how much of the gain is paid out to shareholders. Firm-level differences also translate into unequal wage gains across workers (Alvarez et al., 2018; Song et al., 2019).

This paper aims to address this gap by examining how firms' responses to monetary policy shape its distributional effects through wage and dividend payments. We employ the local projection method of Jordà (2005) to estimate the dynamic responses of firms' revenues, employment, wage payments, and dividend payments to monetary policy shocks. We then examine whether these responses vary with firms' dependence on external finance and assess how the resulting changes in wages and dividends affect the distribution of individual income. To identify monetary policy shocks in Taiwan, we follow the high-frequency identification approach of Bu, Rogers and Wu (2021), which extracts monetary policy surprises from movements in government bond yields around central bank announcement dates. This

approach is particularly well suited to the Taiwanese setting because it can be implemented using daily financial market data and remains informative even during periods in which policy rates are close to the zero lower bound (Brennan et al., 2024).

Our empirical analysis draws on four administrative databases from the Fiscal Information Agency of Taiwan covering the period 2004–2023, which we combine into a firm-level panel and an individual-level panel. On the firm side, corporate income tax and business registration records provide detailed financial statements and firm characteristics for dividend-paying firms, including both listed and unlisted enterprises. On the individual side, personal income tax records report every wage and dividend payment that each individual receives from each firm, and household registration records provide demographic characteristics. Because the personal income tax records identify the firm making each payment, we can link every worker and shareholder to the firms from which they receive income. Such linked firm–worker–shareholder data allow us to trace firms' responses to monetary policy through to the wage and dividend income received by their workers and shareholders, thereby connecting firm-level adjustment to changes in the distribution of individual income.

Taiwan provides a particularly compelling setting for this analysis. In recent years, the Taiwanese economy has been characterised by a striking concentration of growth in the semiconductor and integrated circuit industries, driving strong export performance and robust GDP growth. At the same time, interest rates have remained persistently suppressed, a pattern that *The Economist* has termed the "Formosan flu". While this low interest rate environment has supported export-oriented firms, it has also contributed to surging property prices and stagnant wages in sectors outside the technology industry (The Economist, 2025).

These developments raise a central question: how are the gains from monetary easing divided between workers and shareholders? Wages and dividends are two key channels through which firms pass the gains from monetary easing on to individuals, and these two forms of income accrue to very different parts of the distribution. Wage income is spread broadly across individuals, whereas dividend income is heavily concentrated at the top. If monetary easing leads firms to raise dividends by more than wages, it may widen income inequality even when it raises incomes across the board. Understanding how firms' allocation of earnings between wages and dividends shapes the distributional effects of monetary policy

is therefore central to the growing debate over income inequality in Taiwan. We aim to speak directly to this debate by providing systematic evidence on how firms' wage and dividend responses to monetary policy translate into income gains across the distribution of workers and shareholders.

We find three sets of results. First, expansionary monetary policy raises firms' business revenue and shifts their payments to both workers and shareholders, with dividends responding far more strongly than wages. In our preferred specification, a one-unit expansionary shock, normalised to lower the overnight interbank rate by 12.5 basis points on impact[1] raises firm revenue by 17.56% on impact, and the cumulative response after one year is smaller and statistically insignificant. Employment does not respond significantly. Average wages per employee rise by 3.15% on impact and by 4.06% cumulatively after one year, whereas average dividends per shareholder rise by 13.49% on impact and by 31.51% after one year.

Second, the dividend response is stronger among firms that rely more heavily on short-term borrowing. A one-percentage-point higher pre-shock ratio of short-term loans to total assets raises the impact response of dividends by about 0.09 percentage points. Moving from a firm with no short-term loans to one at the 90th percentile of the short-term loan ratio therefore adds roughly 3.7 percentage points to its dividend response. We find no comparable heterogeneity in revenue, employment, or wages, and the result is robust to measuring borrowing exposure with the overall leverage ratio. These patterns are consistent with a credit channel in which monetary easing relaxes the financing constraints of borrowing-exposed firms and allows them to raise shareholder payouts.

Third, these firm-level responses translate into an unequal distribution of income gains across individuals. Following an expansionary shock, the contribution of wage income to total income growth in the first year rises with pre-shock income rank, from 4.23 percentage points in the bottom decile to 9.32 percentage points in the 90th–99th percentile group, before falling to 7.10 percentage points for the top 1%. The contribution of dividend income, by contrast, is concentrated at the very top. It reaches 11.13 percentage points for the top 1%, compared with between 2.3 and 4.7 percentage points for the other groups. As a result,

[1]The annual shock is the sum of the monthly shocks within each calendar year and therefore retains the same normalisation. Because policy surprises in Taiwan are small, the annual series has a standard deviation of 0.0495 units, so a one-unit shock is considerably larger than the typical annual realisation in our sample.

the implied total income gain of the top 1% (18.23 percentage points) is about 2.2 times that of the bottom decile (8.38 percentage points), and dividends account for about 61% of the top 1%'s gain. Monetary easing in Taiwan thus disproportionately benefits top earners, primarily through dividend income.

This paper makes three contributions to the literature. First, this paper bridges two related but largely separate strands of research and provides an integrated analysis linking firm-level monetary policy transmission to individual income distribution. One strand documents that monetary policy has heterogeneous effects on household income and wealth (Coibion et al., 2017; Mumtaz and Theophilopoulou, 2017; Holm, Paul and Tischbirek, 2021; Amberg et al., 2022; Andersen et al., 2023; Lenza and Slacalek, 2024). A separate strand establishes that firms respond heterogeneously to monetary policy depending on their financial constraints and access to external finance (Bernanke, Gertler and Gilchrist, 1999; Cloyne et al., 2018; Ippolito, Ozdagli and Perez-Orive, 2018; Ottonello and Winberry, 2020; Crouzet, 2021; Thürwächter, 2022; Anderson and Cesa-Bianchi, 2024; Gnewuch and Zhang, 2025). Despite the natural complementarity of these two literatures, studies of household income distribution have largely treated the firm as a black box, while firm-level studies have not traced their results through to the distribution of individual labour and dividend income. The notable exception is Coglianese, Olsson and Patterson (2025), who use population-level administrative data from Sweden to document that monetary tightening raised unemployment disproportionately among low-income workers, and that nominal wage rigidity accounts for approximately half of the overall unemployment response.[2] Our paper extends this line of research by examining how firms divide the gains from monetary easing between wage and dividend payments and how this division shapes inequality in individual income. In doing so, we also relate to the literature on rent sharing, which shows that firms share only a limited fraction of profit gains with their workers (Card et al., 2018; Kline et al., 2019). This literature typically exploits firm-specific profit shocks and focuses on workers' earnings.

[2]Two concurrent working papers also examine firm heterogeneity and worker income in the context of monetary policy. Jasova et al. (2024) use Portuguese administrative data and find that expansionary policy reduces between-firm wage inequality, with small and young firms disproportionately benefiting through the credit channel. Bobasu and Repele (2025) use German matched employer–employee data and find instead that large firms transmit shocks more strongly to wages, attributing this to the greater prevalence of variable pay in larger firms. The divergence in findings across European countries suggests that institutional context plays an important role in shaping the firm channel of monetary policy transmission.

We instead study an aggregate monetary policy shock and trace its effects on both wages and dividends, showing that dividends per shareholder respond several times more strongly than wages per employee. Moreover, we measure firms' exposure to external finance by their short-term borrowing and show that this exposure amplifies the dividend response to monetary easing.

Second, this paper contributes to the literature on dividend payout policy by linking dividend decisions to the credit channel of monetary policy. Existing research has primarily examined dividend policy through the lenses of agency problems, asymmetric information, taxation, and investor preferences, with additional emphasis on dividend smoothing and financial flexibility (Brav et al., 2005; Leary and Michaely, 2011; Farre-Mensa, Michaely and Schmalz, 2014). A related strand shows that payout decisions are closely tied to firms' debt and financing choices (Aivazian, Booth and Cleary, 2006; DeAngelo, Gonçalves and Stulz, 2022; Farre-Mensa, Michaely and Schmalz, 2025). These studies, however, do not consider the role of monetary policy. More recent research connects corporate payout decisions to macroeconomic policy. A notable example is Pazarbaşi (2026), who studies the payout channel of monetary policy and shows that monetary easing reduces precautionary cash demand among cash-rich firms, leading them to increase shareholder payouts primarily through share repurchases. Cohen, Gómez-Puig and Sosvilla-Rivero (2019) also find that the ECB's monetary policies encouraged nonfinancial firms to allocate more resources to shareholder distributions as well as investment. However, neither study centres on how firm leverage shapes the response of cash dividends to monetary policy shocks through the credit channel. This paper addresses this gap by examining leverage-dependent dividend responses to monetary policy shocks, thereby connecting the credit channel to firms' dividend decisions.

Third, to our knowledge this paper is the first to examine the distributional consequences of monetary policy using population-level administrative tax records from Taiwan or any East Asian economy. The existing empirical literature is concentrated almost exclusively in North America and Europe (Holm, Paul and Tischbirek, 2021; Amberg et al., 2022; Andersen et al., 2023; Coglianese, Olsson and Patterson, 2025). Taiwan and East Asian economies offer a distinct institutional context: high household saving rates, industrial structures dominated by export-oriented sectors, and persistently suppressed interest rates that have fuelled asset

price inflation while compressing wage growth outside the technology sector (The Economist, 2025). These features may generate distributional consequences of monetary policy that differ substantially from those documented in the existing literature. Our paper provides the first systematic evidence on these questions using comprehensive administrative records that link firms, workers, and shareholders.

# 2 Taiwanese Monetary Policy Shocks

## 2.1 Identification

We begin our analysis by identifying Taiwanese monetary policy shocks. We follow the high-frequency identification approach of Bu, Rogers and Wu (2021), which extracts policy surprises from movements in government bond yields across different maturities within a daily window surrounding central bank policy announcements, thereby isolating the unexpected component of policy decisions from endogenous responses to macroeconomic conditions. We apply the same framework to domestic monetary policy announcements, using government bond yields at the 2-, 5-, 10-, and 20-year maturities together with repurchase agreement rates at the 30-, 90-, and 180-day horizons.[3] Following Bu, Rogers and Wu (2021), we use the two-year government bond yield as the benchmark instrument.

This method is particularly well-suited to the Taiwanese context for two reasons. First, because it extracts monetary policy surprises from movements in long-term interest rates rather than short-term policy rates, it is better equipped to capture the full stance of monetary policy when short-term rates are constrained near zero. As demonstrated by Brennan et al. (2024), this feature allows the measure to remain informative even during periods of near-zero interest rate environments. This is especially relevant for Taiwan, where short-term interest rates have remained persistently low from 2009 to 2021, making conventional short-rate-based measures less reliable. Second, the identification strategy requires only daily changes in interest rates, rather than intraday tick-level data at the minute frequency typi-

[3]The government bond yields and repurchase agreement rates used to construct the Taiwanese monetary policy shock series are obtained from the Statistical Database of the Central Bank of the Republic of China (Taiwan).

cally needed for standard high-frequency identification approaches. This makes the method directly applicable to the Taiwanese setting, where financial market data are available at the daily frequency but not at finer intraday intervals.

Our final series covers the period from 1 January 2004 to 31 December 2023. The shocks are aggregated to the monthly frequency for the aggregate analysis and to the annual frequency for the micro-level analysis. We normalise the shock series such that a one-unit shock corresponds to a 12.5 basis-point increase in the overnight interbank rate in the first month.[4]

Figure 1 plots the shock series over the sample period from January 2004 to December 2023. The series captures several well-known episodes of monetary easing: following the Global Financial Crisis in 2009 and the COVID-19 pandemic in 2020, the series exhibits multiple consecutive negative values, consistent with the accommodative policy stance adopted by the Central Bank of the Republic of China (Taiwan) during these periods.

## 2.2 Validation: Aggregate Responses

We examine the dynamic effects of monetary policy shocks on six macroeconomic indicators for Taiwan: the overnight interbank rate, the log of seasonally adjusted industrial production, the log of seasonally adjusted consumer price index (CPI), the log of seasonally adjusted imports[5], the log of seasonally adjusted exports, and the log of real effective exchange rate.[6] Each variable is examined separately as the outcome variable in the local projection framework proposed by Jordà (2005). Specifically, for each forecast horizon $h = 0, 1, \ldots, 24$, we estimate the following regression:

$$Y_{t+h} - Y_{t-1} = \alpha^h + \beta^h \left(-\hat{MP}_t\right) + \rho^h X_{t-1} + u_{t+h}, \tag{1}$$

where $Y_{t+h} - Y_{t-1}$ is the long difference of aggregate outcome variable at horizon $h$, $-\hat{MP}_t$ is the expansionary monetary policy shock, and $X_{t-1}$ is the set of control variables.

[4]We normalise to 12.5 basis points rather than the conventional 100 basis points because Taiwan's historical interest rate adjustments have been small in magnitude, and a larger normalisation would produce impulse responses that are difficult to interpret in the Taiwanese context.

[5]Since no official seasonally adjusted series is available for Taiwanese imports, we estimate seasonal patterns by regressing imports on monthly dummy variables while controlling for Lunar New Year effects.

[6]All macroeconomic variables are obtained from the Total Economic Database of the Directorate-General of Budget, Accounting and Statistics (DGBAS), Executive Yuan, Taiwan.

$\beta^h$ captures the cumulative impulse response of the outcome variable to an expansionary monetary policy shock at horizon $h$.

The control variables $X_{t-1}$ include 12 lags of the outcome variable itself and 12 lags[7] of the monetary policy shock. As noted in Section 2.1, the shock series are normalised to a 12.5 basis-point impact on the overnight interbank rate in the first month. The analysis uses monthly data covering the period from January 2004 to December 2023.

Figure 2 presents the impulse response functions of six aggregate macroeconomic variables to an expansionary Taiwanese monetary policy shock that lowers the overnight interbank rate by 12.5 basis points. The blue line represents the estimated impulse response at each horizon, and the shaded region denotes the 90% confidence interval constructed using Newey–West standard errors.

The overnight interbank rate falls by 0.125 percentage points on impact, by construction, and continues to decline to about 0.35 percentage points around months 3 to 4. The decline is statistically significant during the first few months, after which the rate gradually returns toward zero, validating that our estimated shock series captures expansionary monetary policy in Taiwan. Industrial production rises significantly between months 5 and 9, peaking at about 17 log points around month 8, consistent with the standard stimulative effect of easier monetary conditions on real economic activity. The CPI rises gradually, reaching about 1 log point between months 13 and 18. The direction is broadly consistent with the expected inflationary effect of a monetary easing even though the response is not statistically significant at the 90% level. The muted price response may partly reflect Taiwan's energy price policies, including electricity subsidies and a floating mechanism for retail fuel prices that tends to smooth short-term fluctuations in consumer prices.

Imports and exports both rise by roughly 10 to 20 log points between months 6 and 14, although the responses are statistically significant only in some months within this window. While the rise in exports is consistent with improved external competitiveness, the simultaneous expansion in imports reflects Taiwan's role as an export-oriented processing economy: as export demand strengthens, demand for imported intermediate inputs expands

[7]We also tried 24 lags of both outcome variable and monetary policy shocks and the results are robust to additional lags.

accordingly. The real effective exchange rate depreciates after month 12, reaching about 5 to 6 log points by month 24, consistent with the standard interest rate parity mechanism whereby a monetary easing leads to capital outflows and weakens the domestic currency.

**Robustness using Alternative Monetary Policy** In Appendix A, we re-estimate the aggregate impulse responses using an orthogonalised monetary policy surprise constructed following Bauer and Swanson (2023) (hereafter, the alternative shock), normalised to lower the overnight interbank rate by 12.5 basis points as in the baseline. The responses are similar in both shape and magnitude: the overnight interbank rate falls significantly in the first months after the shock, industrial production rises by about 10 log points within the first year, the CPI increases gradually by about 1 log point, imports and exports expand, and the real effective exchange rate depreciates by about 5 log points. The main differences are that the trade and exchange rate responses occur somewhat later.

# 3 Firms' Response to Monetary Policy

We now turn to analyse the impact of monetary policy on firms' annual revenue, employment, and allocation of earnings between wages and dividends. To do this, we draw on four administrative databases obtained from the Fiscal Information Agency of Taiwan, and cover the period from 2004 to 2023 at the annual frequency. The *Profit-Seeking Enterprise Income Tax* database contains firm-level financial statement information, including business revenue, profit, assets, liabilities, short-term borrowing, bank deposits, and cash holdings. The *Business Registration* database provides firms' date of incorporation and their legal structure. The *Individual Income Tax* database records all types of income received by each individual from each firm in a given year, including wage and dividend. Wage consists of salaries and bonuses from employers. Finally, the *Household Registration* database provides individual demographic characteristics, including age and gender. These datasets are linked through the firm identifiers recorded in the individual income tax data.

Our sample construction consists of several steps. We first restrict individuals to those registered in Taiwan and aged between 26 and 65. Each individual's wage and dividend

income is then linked to the firm from which it is received, and firm-level wages and dividends are obtained by aggregating across individuals. Firm-level employment is measured as the number of individuals receiving wage payments from the firm in a given year. Average wage is the firm's total wage payments divided by its number of employees, and average dividend is its total dividend payments divided by the number of individual shareholders receiving dividends from the firm. Because these variables are aggregated from personal income tax records, dividends paid to corporate, institutional, and foreign shareholders, as well as to individuals outside our age range, are not included. Our dividend measure therefore captures the dividend income that firms distribute to domestic individual shareholders, which is the component relevant for the distribution of individual income. Since firms' financial statements are filed on a consolidated basis at the headquarters level, we map each branch to its headquarters and aggregate all firm-level variables at the headquarters level.

The resulting dataset is an unbalanced firm-level panel. All nominal variables are deflated by the GDP deflator to obtain real values.[8] We keep firm-year observations only if the firm pays both positive wages and positive dividends at $t-1$, so that the growth rates of both income components are well defined. Since a large number of firms never distribute dividends, this restriction removes around half of the firm-year observations, and our analysis therefore focuses on how firms with an established payout record adjust the amount of wages and dividends they pay. Finally, to limit the influence of extreme values, we drop observations whose absolute annual growth in outcome variables exceeds 500%. The final sample contains 2,860,046 firm-year observations from 347,469 firms.

Table 1 reports descriptive statistics for the estimation sample. Even when restricting the sample to firms that paid both wages and dividends in the previous year, firm size varies widely. The median firm employs only five workers aged 26 to 65, and at least ten percent of firms have a single such employee, whereas the mean is 19 with a standard deviation of 238.5. Total assets and business revenue display similarly right-skewed distributions: the median firm holds about NT$8.2 million in assets and earns NT$14.0 million in revenue, while the means are many times larger. The sample is therefore dominated by small and

[8] We follow Amberg et al. (2022) who use the GDP deflator rather than the CPI because it better captures the prices faced on the production and income sides.

medium-sized enterprises but also contains a small number of very large firms.

The two payout variables differ markedly in their dispersion. The mean and median of the average wage are close to each other (NT$264.5 thousand and NT$246.5 thousand, respectively), and its coefficient of variation is about 0.7. By contrast, the mean average dividend (NT$377.9 thousand) is roughly twice its median (NT$186.5 thousand), and its coefficient of variation exceeds 3. This pattern is consistent with the view that wages are relatively rigid, whereas dividends are more discretionary, and it motivates our investigation of whether the two forms of payment respond differently to monetary policy.

Turning to financial conditions, the median two-year lagged short-term loan to assets ratio is zero, indicating that more than half of the firms in our sample do not rely on short-term loans. Among firms that do borrow, however, reliance on short-term credit varies considerably: the ratio reaches 9.5 percent at the 75th percentile and 39.4 percent at the 90th percentile. The lagged leverage ratio has a median of 31.7 percent and is spread fairly evenly across its range. Finally, the median lagged net profit margin is 5.3 percent, while its mean and standard deviation are driven by a small number of extreme observations, likely arising from firms with very low revenue.

## 3.1 Empirical Specification

To analyse firms' responses to monetary policy, we apply a local projection framework at the firm level. For each horizon $h = 0, 1$, we estimate:

$$\frac{Y_{i,t+h} - Y_{i,t-1}}{Y_{i,t-1}} = \alpha_i^h + \beta^h(-\hat{MP}_{t-1}) + \rho^h X_{i,t} + \epsilon_{i,t}^h, \quad h = 0, 1. \tag{2}$$

where $Y_{i,t}$ denotes one of four firm-level outcomes: annual business revenue, employment, average wage per employee, and average dividend per shareholder. The dependent variable is thus the cumulative growth rate of outcome $Y_i$ for firm $i$ between year $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$. $\alpha_i^h$ is a firm fixed effect, which absorbs all time-invariant firm-level heterogeneity. $\beta^h$ therefore captures the cumulative impulse response of $Y_i$ to an expansionary shock from $t-1$ to $t+h$, identified from within-firm variation. $X_{i,t}$ is a vector of controls: (i) firm age fixed effects, motivated by

Cloyne et al. (2018) who document that firm age shapes firms' behavioural responses; (ii) the firm's net profit margin lagged two periods so that it is determined prior to the realisation of monetary policy; (iii) dynamic controls namely two lags of the dependent variable[9] and one additional lag of the monetary policy shock at $t-2$; and (iv) two extended lags for monetary policy shock at $t-3$ and $t-4$. Finally, $\epsilon^h$ is the error term. Standard errors are two-way clustered at the firm and year levels to account for within-firm serial correlation and cross-sectional dependence within each year.

Our specification follows the literature using shocks constructed via Bu, Rogers and Wu (2021), such as Di Giovanni and Rogers (2024), who similarly employ a lagged rather than contemporaneous monetary policy shock. This choice reflects a timing issue inherent in aggregating high-frequency monetary surprises to annual frequency: since the Bu, Rogers and Wu (2021) shock series is constructed from interest-rate changes around each policy meeting within the year and then summed into an annual measure, a shock realised at the final meeting of year $t$ (e.g., December) has had almost no time to transmit to firm outcomes within that same calendar year. Using the shock realised in year $t-1$ and measuring the cumulative growth of outcomes between year $t-1$ and year $t+h$ therefore ensures that the outcomes we compare against the pre-shock level, from year $t$ onward, have been exposed to the shock for at least a full year.

## 3.2 Results

Tables 2–5 report firms' average responses to an expansionary monetary policy shock equivalent to a 12.5-bp reduction in the overnight interbank rate. All tables contain two panels. Panel A reports the estimates for $h=0$, and Panel B reports those for $h=1$. Each table presents five specifications: Column (1) includes firm fixed effects. Column (2) adds firm age fixed effects, and column (3) further includes the firm's net profit margin lagged two periods, so that it is determined prior to the realisation of monetary policy. Column (4) adds dynamic controls, including two lags of the dependent variable and one additional lag of the monetary policy shock at $t-2$. Column (5) further includes two extended lags of the

[9] Specifically, we include $\frac{Y_{i,t-1}-Y_{i,t-2}}{Y_{i,t-2}}$ and $\frac{Y_{i,t-2}-Y_{i,t-3}}{Y_{i,t-3}}$.

monetary policy shock at $t-3$ and $t-4$. Standard errors are two-way clustered at the firm and year levels to account for within-firm serial correlation and cross-sectional dependence within each year.

Table 2 reports firms' business revenue responses to expansionary monetary policy. Panel A shows a positive and statistically significant response at $h = 0$ across all five specifications, with estimates exceeding 15%. In Panel B, the cumulative response at $h = 1$ remains positive, with point estimates of at least 7.5% even though they are statistically insignificant. In our preferred specification, column (5), the estimated revenue increases by 17.56% at $h = 0$ and the cumulative response is 9.515% at $h = 1$. The positive revenue response and its pattern are consistent with the increase in aggregate industrial production documented in the previous section and suggest that stronger aggregate demand may contribute to higher firm revenue following an expansionary monetary policy shock.

Table 3 reports how firms' employment responds to expansionary monetary policy. In Panel A, employment rises by an estimated 2.683% to 5.998% at $h = 0$. The estimates are statistically significant in columns (1) through (4), but not in our preferred specification, column (5). In Panel B, the cumulative response at $h = 1$ is statistically insignificant in columns (2) through (5); only column (1) is significant at the 10% level. In column (5), the estimated employment response is 2.683% at $h = 0$ and $-0.456\%$ at $h = 1$, neither of which is statistically distinguishable from zero. Taken together, the preferred estimates provide limited evidence that the policy shock leads firms to expand employment.

Table 4 reports how firms' average wage per employee responds to expansionary monetary policy. As Panel A shows, average wages rise by more than 3% at $h = 0$ across all five specifications, with every estimate statistically significant. Panel B shows that wages remain elevated at $h = 1$: the estimated cumulative increases exceed 4% and remain statistically significant in all five specifications. In our preferred specification, column (5), average wages rise by an estimated 3.147% at $h = 0$, while the cumulative increase reaches 4.057% at $h = 1$. These findings are consistent with firms raising wages as their revenue expands following the policy shock. Unlike the revenue response, the estimated wage response shows no sign of fading by $h = 1$.

Table 5 reports how average dividends per shareholder respond to expansionary monetary

policy. As Panel A shows, average dividends rise by an estimated 9.5% or more at $h = 0$ across the five specifications, although the estimates are statistically significant only in columns (1), (4), and (5). By $h = 1$, the estimated cumulative increase exceeds 25% in every specification, and all five estimates are statistically significant at the 1% level. In our preferred specification, column (5), average dividends rise by an estimated 13.49% at $h = 0$, and the cumulative increase reaches 31.51% at $h = 1$. These estimates are considerably larger than the corresponding increases in average wages, suggesting that dividend income responds more strongly to the expansionary shock.

The increase in firms' real revenue is consistent with the rise in aggregate industrial production documented earlier. Both responses are strongest in the short run and diminish thereafter. Our labour-market results are less uniform. Average wages rise and remain elevated at $h = 1$, whereas the employment response is not statistically significant in our preferred specification. This distinction between wage and employment responses is relevant to studies that examine the intensive and extensive margins of labour income (Hubert and Savignac, 2024). The dividend response is more pronounced. Average dividends rise by substantially more than average wages in our preferred specification, particularly at $h = 1$. The increase in dividends is consistent with studies of payouts. For example, Cohen, Gómez-Puig and Sosvilla-Rivero (2019) document that easier ECB policy leads to higher leverage and shareholder yield. Daniel, Garlappi and Xiao (2021) predict that low interest rates may induce firms to initiate dividends to attract income-seeking investors. They also present suggestive evidence that dividend initiations are more common when interest rates are low. Our results add evidence on the amount of dividend income paid per shareholder after an expansionary shock. This distinction between wages and dividends also matters for the distribution of individual income. Amberg et al. (2022) find that capital income accounts for much of the monetary-policy response at the top of the income distribution, while Andersen et al. (2023) document particularly large gains in stock market income among high-income individuals. The larger estimated response of dividends than wages in our results suggests that monetary policy may affect the two sources of income differently. We examine the distributional implications directly in Section 5 using individual-level income data.

**Robustness using Alternative Monetary Policy** In Appendix A, we show the results using the alternative shock. The results are qualitatively similar to our baseline. Revenue rises significantly on impact and the effect diminishes by $h = 1$, average wages rise and remain elevated at $h = 1$, and average dividends increase by considerably more than wages at $h = 1$. The main differences lie in magnitude. The revenue and dividend responses are roughly half as large as in the baseline, whereas the wage response is nearly identical. In addition, the employment response at $h = 0$ becomes statistically significant, although it remains modest and does not persist at $h = 1$.

**Robustness using Alternative Deflator** In Appendix B, we examine whether our findings depend on the choice of price deflator by deflating all nominal variables by the CPI instead of the GDP deflator. Because the deflator is common to all firms in a given year, this change shifts the estimated real growth rates of revenue, wages, and dividends by approximately the same amount. Under the CPI, the revenue and dividend responses remain positive and statistically significant, although somewhat smaller, whereas the average wage response is no longer statistically distinguishable from zero. This indicates that firms raise wages relative to output prices, but that nominal wage growth roughly keeps pace with consumer prices. Importantly, the gap between the dividend and wage responses is virtually unchanged, so our central finding that dividend income responds considerably more strongly than wage income is robust to the choice of deflator.

# 4 The Credit Channel and Dividend Payouts

Our baseline results show that expansionary monetary policy raises both average wages and average dividends, with a substantially larger estimated percentage increase in dividends. We next examine whether firms' borrowing exposure helps explain this strong dividend response. The mechanism we have in mind involves two links. First, monetary easing should benefit firms with more short-term borrowing disproportionately, because such debt is repriced or refinanced quickly. Consistent with this, existing research shows that the transmission of monetary policy depends on firms' debt structure, particularly their reliance on bank loans

(Ippolito, Ozdagli and Perez-Orive, 2018; Holm-Hadulla and Thürwächter, 2021; Crouzet, 2021). Second, the resulting relief in financing costs may be passed on to shareholders, as borrowing and payout decisions are closely linked (Aivazian, Booth and Cleary, 2006; DeAngelo, Gonçalves and Stulz, 2022; Farre-Mensa, Michaely and Schmalz, 2025). While each link has been documented separately, existing studies do not examine whether monetary policy, through firms' borrowing exposure, shapes dividend payments relative to wages.

To test this mechanism, we measure each firm's pre-existing borrowing exposure as the ratio of short-term loans to total assets at $t-2$, which we refer to as the short-term loan ratio.[10] We then interact this ratio with the monetary policy shock to test whether firms with greater short-term borrowing increase dividends more strongly following an expansionary shock. We estimate the same interaction for average wages to assess whether the pattern also appears in payments to workers:

$$\begin{aligned}\frac{Y_{i,t+h}-Y_{i,t-1}}{Y_{i,t-1}} = \alpha_i^h + \beta_1^h(-\hat{MP}_{t-1}) + \beta_2^h\, Exposure_{i,t-2} \\ + \gamma^h\, Exposure_{i,t-2}(-\hat{MP}_{t-1}) + \rho^h X_{i,t} + \epsilon_{i,t}^h, \quad h=0,1.\end{aligned} \tag{3}$$

$Exposure_{i,t-2}$ is the short-term loan ratio lagged two periods for the same reason as the net profit margin control described above: by construction, it is determined prior to the realisation of $-\hat{MP}_{t-1}$, ruling out reverse causality between the shock and firms' financing needs. $\gamma^h$ is our coefficient of interest: a positive and significant $\gamma^h$ would indicate that firms with greater exposure to short-term credit increase dividend payouts by more in response to expansionary monetary policy, consistent with a credit channel through which monetary easing relaxes the financing constraints of borrowing-exposed firms. $X_{i,t}$ retains the same set of controls as in equation (2).

Tables 6–9 examine whether firms' responses to an expansionary monetary policy shock vary with their short-term loans relative to total assets, measured at $t-2$. The four tables report results for business revenue, employment, average wages per employee, and average dividends per shareholder, respectively. Panel A reports estimates for $h=0$, while Panel

[10] Our scaling is similar to Ippolito, Ozdagli and Perez-Orive (2018), who measure bank debt relative to total assets. Appendix C reports results using leverage ratio (total liabilities divided by total assets) and shows similar patterns.

B reports estimates for $h = 1$. All specifications include the monetary policy shock, the short-term loan ratio, and their interaction. Column (1) includes firm fixed effects; column (2) adds firm age fixed effects; and column (3) further controls for net profit margin at $t-2$. Column (4) adds two lags of the outcome's growth rate and the monetary policy shock at $t-2$. Column (5) further includes the policy shocks at $t-3$ and $t-4$. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses.

Table 6 examines whether the revenue response varies with firms' short-term loan ratios. In Panel A, the interaction estimates are positive across all five specifications, but none is statistically significant. The same holds in Panel B. In our preferred specification, column (5), a one-percentage-point higher short-term loan ratio is associated with an additional 0.0128-percentage-point revenue response at $h = 0$ and an additional 0.00626-percentage-point cumulative response at $h = 1$. Neither estimate provides evidence that revenue responds differently across firms with different short-term loan ratios.

Table 7 reports the corresponding results for employment. The interaction estimates are positive but statistically insignificant in all five specifications at both horizons. In column (5), a one-percentage-point higher short-term loan ratio is associated with an additional 0.0232-percentage-point employment response at $h = 0$ and an additional 0.0297-percentage-point cumulative response at $h = 1$. We therefore find no statistically significant difference in employment responses by short-term loan exposure.

Table 8 turns to average wages per employee. Unlike the revenue and employment estimates, the interaction estimates are negative at both horizons, but none is statistically significant. In column (5), a one-percentage-point higher short-term loan ratio is associated with a 0.0126-percentage-point smaller wage response at $h = 0$ and a 0.00634-percentage-point smaller cumulative response at $h = 1$. These estimates provide no evidence that firms with greater short-term borrowing raise average wages more strongly following the policy shock.

Table 9 reports the results for average dividends per shareholder. In Panel A, the interaction estimates are positive and statistically significant across all five specifications. In our preferred specification, column (5), a one-percentage-point higher short-term loan ratio increases the estimated dividend response at $h = 0$ by 0.0930 percentage points. In Panel B,

the interaction estimates remain positive but are no longer statistically significant; the column (5) estimate is 0.0474 percentage points. Thus, firms with greater short-term borrowing exhibit a stronger initial dividend response, while the evidence of a differential response does not persist at $h = 1$.

The interaction results show that firms with higher pre-existing short-term loan ratios increase average dividends more strongly following an expansionary monetary policy shock. In our preferred specification, a one-percentage-point higher ratio is associated with an additional 0.0930-percentage-point dividend response at $h = 0$. We do not find a statistically significant interaction for wages, employment, or revenue, and the dividend interaction is no longer significant at $h = 1$. This pattern is consistent with a financing channel: easing may reduce debt-servicing or refinancing costs for firms exposed to short-term borrowing, giving them greater flexibility to distribute funds to shareholders. Because we do not directly observe the change in borrowing costs, we interpret this as a possible mechanism rather than a demonstrated intermediate effect.

Investor demand may also influence payout decisions. Daniel, Garlappi and Xiao (2021) show that lower interest rates increase demand for high-dividend stocks and provide suggestive evidence that firms are more likely to initiate cash dividends when interest rates are low. Greater financial flexibility could allow borrowing-exposed firms to respond to such demand, although our estimates do not identify a catering motive directly.

**Robustness using Alternative Monetary Policy** In Appendix A, we show the results using the alternative shock. Because this change alters the variation used to identify the interaction coefficient, it provides an independent check on our cross-sectional results. The pattern is qualitatively unchanged. Firms with higher short-term loan ratios again exhibit a significantly stronger dividend response at $h = 0$, and this differential response does not persist at $h = 1$. The interactions for revenue, employment, and wages remain statistically insignificant. The main difference lies in magnitude: in our preferred specification, the dividend interaction at $h = 0$ is 0.0565 percentage points, roughly 60% of the baseline estimate of 0.0930, consistent with the smaller average dividend response under this shock.

**Robustness using Alternative Deflator** In Appendix B, we again examine whether our findings depend on the choice of price deflator by deflating all nominal variables by the CPI instead of the GDP deflator. Because the deflator is common to all firms in a given year, this change mainly affects the average response to the shock rather than the differential response across firms with different short-term loan ratios. Consistent with this, the interaction estimates are nearly identical to the baseline. Firms with higher short-term loan ratios again exhibit a significantly stronger dividend response at $h = 0$, with a column (5) estimate of 0.0922 percentage points compared with 0.0930 in the baseline, and this differential response does not persist at $h = 1$. We find no significant interaction for revenue or wages under either deflator. The evidence that short-term borrowing exposure amplifies the dividend response to expansionary monetary policy is therefore robust to the choice of price index.

**Alternative Measure of Credit Exposure** In Appendix C, we examine whether these results depend on how credit exposure is measured by replacing the short-term loan ratio with the leverage ratio, defined as total liabilities divided by total assets. The leverage ratio is a broader measure of reliance on external financing, as it also includes long-term debt and non-bank liabilities that are less directly affected by short-term policy rates. The results closely mirror our baseline findings. Firms with higher leverage exhibit a significantly stronger dividend response at $h = 0$, with a column (5) estimate of 0.0734 percentage points compared with 0.0930 for the short-term loan ratio, while the interactions for revenue, employment, and wages remain statistically insignificant. The somewhat smaller dividend interaction is consistent with the broader leverage measure capturing liabilities that are less sensitive to monetary policy.

# 5 Distributional Implications for Individual Income

The firm-level results show that monetary easing raises dividends considerably more than wages. Because dividend income is far more concentrated than wage income, this asymmetry may also be reflected in the distribution of individual income. We therefore examine whether

the effects of monetary policy on income accrue disproportionately to top earners, who are more likely to hold dividend-paying assets. This exercise also allows us to compare our findings with existing evidence from Northern Europe, which shows that the income gains from monetary easing at the top of the distribution are driven largely by capital income (Holm, Paul and Tischbirek, 2021; Amberg et al., 2022; Andersen et al., 2023).

We use the same individual records used to construct the firm-level variables, restricting the sample to employees and shareholders aged 26–65, and apply the same restriction excluding absolute changes exceeding 500% to eliminate extreme values. We decompose each individual's pre-tax income into wage income and dividend income, and rank individuals into eleven groups based on their total income (wage plus dividend) at $t-2$: the bottom nine deciles (the 0th–10th, 10th–20th, ..., 80th–90th percentiles), the 90th–99th percentiles, and the top 1%.

Table 10 reports summary statistics for individuals grouped by their total income at $t-2$. Three patterns stand out. First, wage income at $t$ rises steadily with income rank, from NT$0.26 million in the bottom decile to NT$4.09 million in the top 1%, a ratio of roughly 16. Second, dividend income is far more unequally distributed. Average dividends at $t$ are relatively flat across the bottom half of the distribution, ranging between NT$42,000 and NT$63,000, but increase sharply at the top, reaching NT$5.78 million for the top 1%, about 138 times the level of the bottom decile. Third, dividends are a much more important source of income at the top of the distribution. At $t-2$, dividends account on average for 10 to 14 percent of an individual's total income in the bottom nine deciles, compared with 20 percent in the 90th–99th percentile group and 46 percent in the top 1%. Because this ratio is measured before the realisation of the monetary policy shock, it captures each group's pre-existing exposure to changes in dividend payouts. By contrast, age varies little across most groups, staying between 42 and 46 years for the bottom nine deciles.

Because the individual sample is drawn from the final firm-level sample, it only includes individuals who receive income from firms that pay both wages and dividends. The income composition reported in Table 10 should therefore be interpreted as describing income received from dividend-paying firms, rather than individuals' total income or the income distribution of the population as a whole.

We then estimate the following specification:

$$\frac{Y^k_{i,t+h} - Y^k_{i,t-1}}{Y_{i,t-1}} = \sum_{g=1}^{11} G_{i,t-2,g} \times \left[\alpha^h_g + \beta^h_g(-\hat{MP}_{t-1})\right] + \rho^h X_{i,t} + \epsilon^h_{i,t}, \quad h = 0, 1. \tag{4}$$

This specification is a group-specific local projection model. The dependent variable is not a simple growth rate, but a *growth-contribution* measure, following Amberg et al. (2022): the change in income component $Y^k_i$ (wage or dividend income) between $t-1$ and $t+h$, scaled by individual $i$'s *total* pre-tax income at $t-1$. Since the wage and dividend contribution terms sum to individual $i$'s overall income growth rate, $\beta^h_g$ can be interpreted as the percentage points of total income growth attributable to component $Y^k$ for group $g$, $h$ years after an expansionary shock. $G_{i,t-2,g} = \mathbb{1}[\text{IncomeRank}_{i,t-2} = g]$ is a binary indicator equal to one if individual $i$'s total income rank at $t-2$ places them in percentile group $g$; income groups are assigned prior to the realisation of $-\hat{MP}_{t-1}$ to ensure they are not mechanically affected by the shock itself. $-\hat{MP}_{t-1}$ is the lagged expansionary Taiwan monetary policy shock, as defined above. $\alpha^h_g$ and $\beta^h_g$ are group-specific intercepts and impulse responses, respectively, allowing the dynamic response of income to monetary policy to vary flexibly across the income distribution. $X_{i,t}$ collects both individual and age-group fixed effects jointly.

Figure 3 shows the individual income response to expansionary monetary policy. The four panels present the estimated contributions of wage and dividend income to total income growth across the eleven pre-shock income groups. Panels (a) and (b) show the responses at $h = 0$, and panels (c) and (d) show those at $h = 1$. Panels (a) and (c) report the wage responses, and panels (b) and (d) report the dividend responses. In each panel, the horizontal axis runs from the bottom 10% of the income distribution to the top 1%, and the vertical axis reports the income component's contribution to total income growth in percentage points. The bars show 95% confidence intervals based on standard errors clustered at the individual and year levels.

At $h = 0$, the wage contribution increases steadily with pre-shock income rank, from 4.227 percentage points in the bottom decile to 9.321 percentage points in the 90th–99th percentile group, before falling to 7.098 percentage points for the top 1%. All estimates are significant

at the 5% level or better. At $h = 1$, this gradient disappears. The estimates lie between 5.140 and 5.721 percentage points for all groups below the top 1% and are significant at the 5% level for the bottom nine deciles, but not for the 90th–99th percentile group (5.721). The estimate for the top 1% is lower, at 4.737 percentage points, and is also statistically insignificant. The upward income gradient in the wage response is thus concentrated in the year of the shock.

The dividend contribution follows a different pattern. At $h = 0$, it is 4.152 and 4.685 percentage points for the bottom two deciles, declines gradually to 2.280 percentage points for the 80th–90th percentile, rises to 3.845 percentage points for the 90th–99th percentile group, and jumps to 11.13 percentage points for the top 1%. All estimates are significant at the 10% level or better. The top 1% contribution is about 2.7 times that of the bottom decile and 2.9 times that of the 90th–99th percentile group. Unlike the wage response, the dividend response persists: at $h = 1$, the contributions are generally larger than at $h = 0$ and remain significant for all groups, reaching 13.06 percentage points for the top 1%, significant at the 10% level.

Because the wage and dividend contributions sum to total income growth by construction, adding the two point estimates gives the implied total response.[11] At $h = 0$, the implied total is 8.38 percentage points in the bottom decile. It remains between 9.6 and 10.0 percentage points for the 10th–70th percentiles. It then rises to 13.17 percentage points for the 90th–99th percentile group and reaches 18.23 percentage points for the top 1%. The response of the top 1% is therefore about 2.2 times that of the bottom decile. Moreover, the increase from the 90th–99th percentile group to the top 1% (5.06 percentage points) exceeds the entire increase from the bottom decile to the 90th–99th percentile group (4.79 percentage points).

This jump at the very top is driven by dividend income. The wage contribution of the top 1% is smaller than that of the 90th–99th percentile group, while dividends account for about 61% of the top 1%'s implied total response. At $h = 1$, the implied total lies between 8.8 and 10.8 percentage points for all groups below the top 1%. For the top 1%, it is 17.80 percentage points. The concentration of gains at the top therefore persists into the following

[11] The implied totals are sums of point estimates from separate regressions and are therefore approximate.

year.

Taken together, these patterns suggest that the income gains following expansionary monetary policy are tilted toward higher-income individuals, and that the two income components contribute to this tilt in different parts of the distribution. The wage contribution increases with income rank up to the 90th–99th percentile but not for the top 1%, whereas the dividend contribution is distinctly large and persistent for the top 1%. The combination of these two patterns produces a total effect that rises across the upper half of the distribution and peaks at the very top.

The pattern of dividend contributions mirrors the pre-shock importance of dividends in individuals' income. As reported in Table 10, dividends account on average for 10–14% of total income at $t-2$ for the bottom nine deciles, 20% for the 90th–99th percentile group, and 46% for the top 1%. When firms raise dividend payouts following monetary easing, as shown in our firm-level results, the effect on total income is therefore largest for the top 1%, for whom dividends constitute by far the largest share of income. The relatively large dividend contributions in the bottom two deciles should be interpreted with caution: because the contribution is scaled by total income, which is low for these groups, even small absolute changes in dividends translate into sizeable contributions. For the middle and upper-middle of the distribution, wage income is the main channel through which monetary easing affects income.

The concentration of dividend gains at the top is consistent with evidence that capital-income responses to monetary policy are strongest among high-income individuals (Amberg et al., 2022) and that gains from monetary easing increase with prior income (Andersen et al., 2023). Our wage-income pattern, however, differs from the U-shaped response documented by Amberg et al. (2022), whose large response at the bottom is driven by labour income; in our data, the relatively large response at the bottom comes from dividends rather than wages. This difference may partly reflect our sample restriction to firms that pay both wages and dividends. By isolating dividends, our analysis identifies one specific source of the top-income response. It cannot capture responses in realised capital gains or other forms of capital income.

**Robustness using Alternative Monetary Policy** In Appendix A, we show the results using the alternative shock. Using this alternative shock, we obtain the same distributional patterns as in the baseline in the year of the shock: the wage contribution rises with income rank up to the 90th–99th percentile but not for the top 1%, the dividend contribution is distinctly large for the top 1%, and the implied total income gain peaks at the very top. The main difference is that the persistence of the top-income response in the following year is weaker and imprecisely estimated. Overall, our conclusion that expansionary monetary policy tilts income gains toward top earners, largely through dividends, is not driven by the choice of monetary policy shock.

**Robustness using Alternative Deflator** In Appendix B, we deflate wage and dividend income by the CPI instead of the GDP deflator. The dividend contributions are almost identical to the baseline: at $h = 0$, the top 1% contribution is 11.20 percentage points, compared with 11.13 percentage points in the baseline, and it remains large and significant at $h = 1$. The wage contributions are smaller across the distribution and are not significant at $h = 1$, but they still rise with income rank up to the 90th–99th percentile and are lower for the top 1%. As a result, the implied total income gain again peaks at the very top in both years, with dividends accounting for an even larger share of the top 1%'s response. Our distributional conclusions are therefore robust to the choice of price deflator.

# 6 Conclusion

This paper examines how firms' allocation of earnings between wages and dividends shapes the distributional effects of monetary policy in Taiwan. Combining high-frequency monetary policy shocks with administrative tax records that link firms to their workers and shareholders over 2004–2023, we document three findings. First, expansionary monetary policy raises firm revenue and increases both average wages and average dividends, but dividends respond far more strongly: in our preferred specification, average dividends per shareholder rise by about 31.5% after one year, compared with about 4.1% for average wages. Second, the initial dividend response is larger among firms with greater pre-existing short-term borrow-

ing, whereas we find no comparable heterogeneity in revenue, employment, or wages. This pattern is consistent with a credit channel operating through shareholder payouts. Third, because dividend income is heavily concentrated at the top, the stronger dividend response translates into unequal income gains across individuals: the implied total income gain of the top 1% is about 2.2 times that of the bottom decile, and dividends account for about 61% of the top 1%'s gain. These patterns are robust to an alternative identification of monetary policy shocks, to deflating by the CPI, and to measuring credit exposure with the leverage ratio.

Our analysis has several limitations that point to directions for future research. Our estimates are reduced-form and do not identify why borrowing-exposed firms raise dividends rather than wages. Our data do not allow us to distinguish whether monetary easing frees up cash flow through lower debt-servicing costs, allows firms to finance payouts with new debt, or leads firms to cater to income-seeking investors. Moreover, the stronger response of dividends relative to wages may reflect differences in adjustment flexibility. Dividends are residual claims that firms can adjust from year to year, whereas wages are subject to contractual and nominal rigidities. Wages may therefore respond more slowly than dividends. We cannot, however, test this mechanism directly. In addition, although our sample covers all publicly listed firms that pay dividends, most firms by count are small and medium-sized enterprises, many of which are closely held. For these firms, the division between wages and dividends may partly reflect owner-managers' compensation and tax choices. Finally, our individual-level analysis shows that the dividend channel tilts income gains toward top earners, but does not directly trace this tilt to firms' borrowing exposure. A model of firms' earnings allocation that incorporates financing frictions and wage setting, together with tests that exploit the overlap between workers and shareholders in our linked data, would help clarify how the gains from monetary easing are divided between labour and capital.

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# Figure

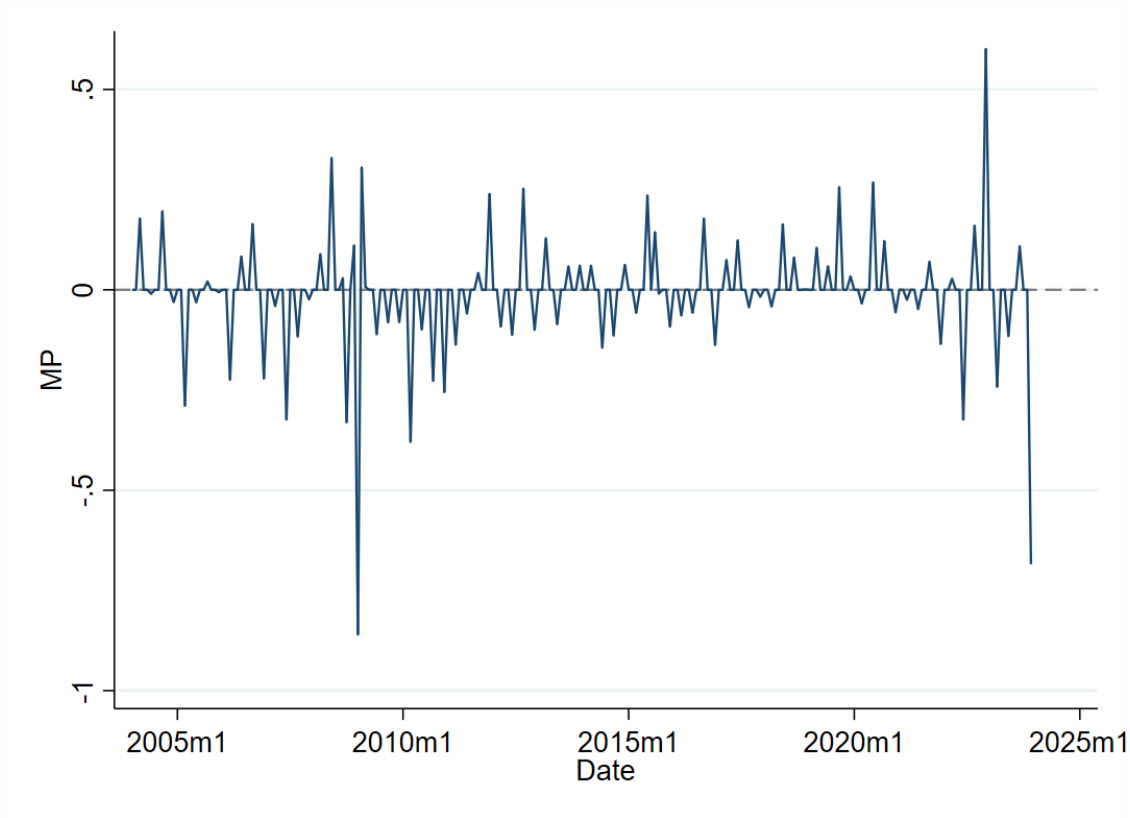


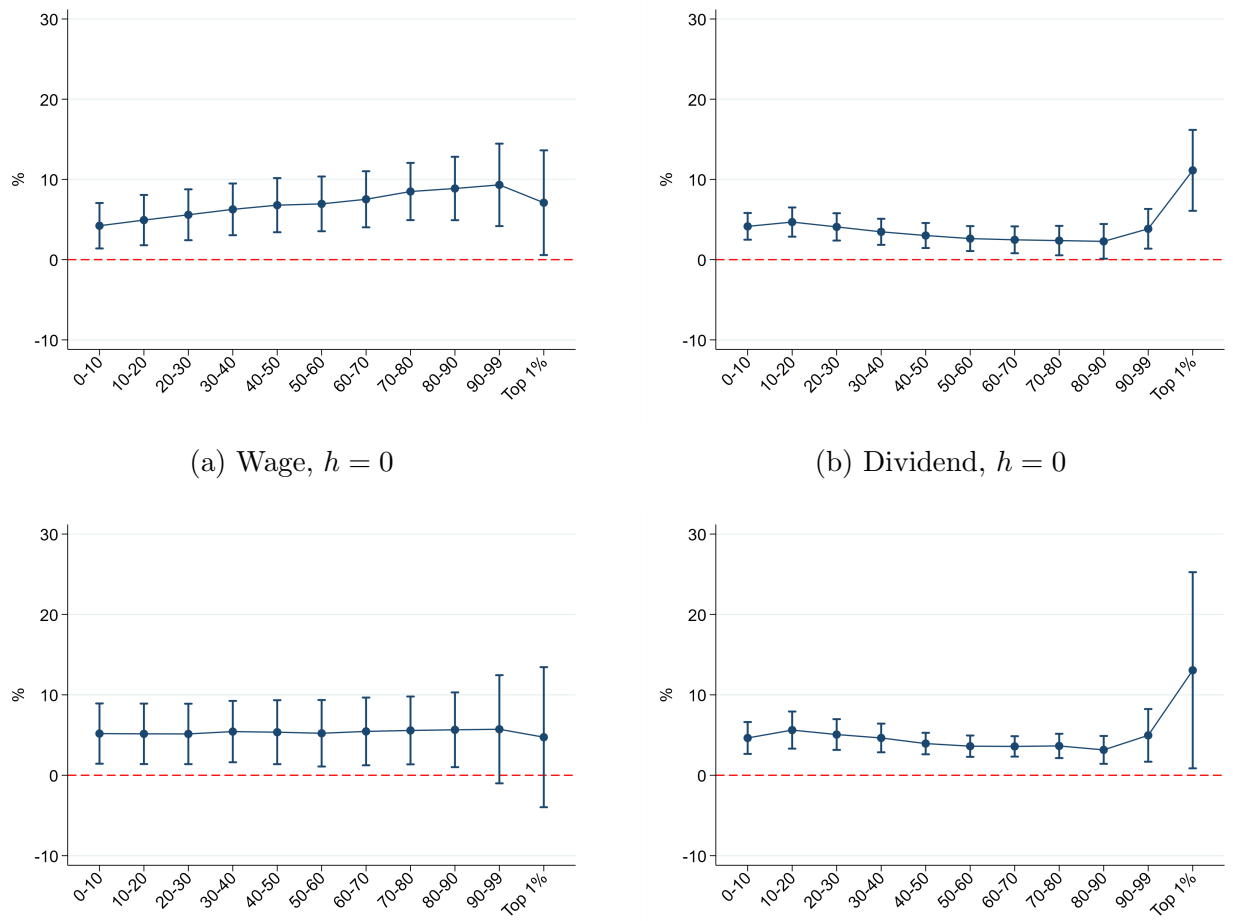

Figure 1: Estimated Taiwanese Monetary Policy Shocks Series

*Note:* This figure plots the Taiwanese monetary policy shock series from January 2004 to December 2023, aggregated to the monthly frequency. The shocks are identified following the high-frequency approach of Bu, Rogers and Wu (2021), using daily changes in the 2-, 5-, 10-, and 20-year government bond yields and the 30-, 90-, and 180-day repurchase agreement rates around the policy announcements of the Central Bank of the Republic of China (Taiwan), with the two-year government bond yield as the benchmark instrument. The series is normalised so that a one-unit shock raises the overnight interbank rate by 12.5 basis points in the first month. Positive values therefore indicate contractionary surprises, and negative values indicate expansionary surprises. Interest rate data are from the Statistical Database of the Central Bank of the Republic of China (Taiwan).




Figure 2: Impulse Responses to Taiwanese Monetary Policy Shocks

*Note:* This figure shows the cumulative impulse responses of six Taiwanese macroeconomic variables to an expansionary monetary policy shock that lowers the overnight interbank rate by 12.5 basis points in the first month, estimated from the local projection in Equation (1) using monthly data from January 2004 to December 2023. The outcome variables are the overnight interbank rate, in percentage points, and the logs of seasonally adjusted industrial production, the seasonally adjusted consumer price index, seasonally adjusted imports, seasonally adjusted exports, and the real effective exchange rate, in log points. Each regression controls for 12 lags of the outcome variable and 12 lags of the monetary policy shock. The horizontal axis shows the number of months after the shock. Solid blue lines denote point estimates, and shaded areas denote 90% confidence intervals based on Newey–West standard errors. Macroeconomic data are from the Directorate-General of Budget, Accounting and Statistics (DGBAS), Executive Yuan, Taiwan.




(a) Wage, $h = 0$

(b) Dividend, $h = 0$

(c) Wage, $h = 1$

(d) Dividend, $h = 1$

Figure 3: Individual Impulse Responses of Wage and Dividend

*Note:* This figure shows the estimated contributions of wage income (panels (a) and (c)) and dividend income (panels (b) and (d)) to individual total income growth following an expansionary monetary policy shock that lowers the overnight interbank rate by 12.5 basis points, at $h = 0$ (panels (a) and (b)) and $h = 1$ (panels (c) and (d)), estimated from Equation (4). The dependent variable is the change in each income component between $t - 1$ and $t + h$, divided by the individual's total income (wage plus dividend income) at $t - 1$. The estimates are therefore expressed in percentage points of total income growth, and the wage and dividend contributions sum to total income growth by construction. The two contributions are estimated in separate regressions. Individuals are grouped by their total income rank at $t - 2$ into the bottom nine deciles, the 90th–99th percentiles, and the top 1%. In each panel, the horizontal axis lists the eleven income groups and the vertical axis reports the contribution in percentage points. All regressions include individual and age-group fixed effects, and all income is deflated by the GDP deflator. The sample consists of individuals aged 26–65 who receive wage or dividend income from firms in the firm-level sample; observations with absolute changes exceeding 500% are excluded. Dots denote point estimates, and bars denote 95% confidence intervals based on standard errors two-way clustered at the individual and year levels.

# Table

Table 1: Firm-Level Descriptive Statistics

| | Mean | Standard deviation | 10th percentile | 25th percentile | 50th percentile | 75th percentile | 90th percentile |
|---|---|---|---|---|---|---|---|
| **Panel A: Dependent variables** | | | | | | | |
| Average Wage | 264.5 | 186.2 | 73.9 | 159.8 | 246.5 | 334.0 | 455.4 |
| Average Dividend | 377.9 | 1,186.5 | 15.1 | 68.5 | 186.5 | 426.5 | 834.8 |
| **Panel B: Financial conditions** | | | | | | | |
| Short-Term Loan Ratio$_{t-2}$ (%) | 9.35 | 18.21 | 0.00 | 0.00 | 0.00 | 9.51 | 39.36 |
| Leverage Ratio$_{t-2}$ (%) | 35.96 | 29.65 | 1.56 | 6.08 | 31.74 | 63.08 | 78.58 |
| Net Profit Margin$_{t-2}$ (%) | 87.10 | 40,493.64 | 1.47 | 3.54 | 5.33 | 6.00 | 7.42 |
| **Panel C: Other firm characteristics** | | | | | | | |
| Number of Employees | 19.0 | 238.5 | 1 | 2 | 5 | 11 | 27 |
| Business Revenue | 245,282.8 | 13,764,255.8 | 2,262.4 | 5,430.7 | 14,024.9 | 33,981.3 | 103,178.6 |
| Assets | 552,679.4 | 33,304,792.8 | 1,552.8 | 3,450.5 | 8,198.2 | 24,393.3 | 82,141.8 |
| Short Loan | 222,125.3 | 23,796,892.7 | 0.0 | 0.0 | 0.0 | 1,981.8 | 13,463.1 |
| Total Profit | 15,695.5 | 972,952.2 | 39.0 | 192.8 | 556.0 | 1,374.3 | 3,535.6 |
| Observations | 2,860,046 | | | | | | |

*Notes:* This table reports descriptive statistics for the estimation sample described in Section 3, which consists of 2,860,046 firm-year observations covering 347,469 firms from 2004 to 2023. Data are from the Fiscal Information Agency of Taiwan. The sample includes firms with positive wage and dividend payments in year $t-1$, and excludes observations with absolute annual growth rates exceeding 500 percent. All monetary variables are deflated by the GDP deflator and expressed in thousands of New Taiwan dollars. Number of Employees is the number of individuals aged 26 to 65 receiving wage payments from the firm. Average Wage is total wage payments, including salaries and bonuses, divided by the number of employees. Average Dividend is total dividend payments divided by the number of dividend recipients aged 26 to 65. Short-Term Loans are short-term borrowings reported on the firm's balance sheet. Short-Term Loan Ratio$_{t-2}$ is short-term loans divided by total assets. Leverage Ratio is total liabilities divided by total assets. Net Profit Margin is net profit divided by business revenue. All ratios are expressed in percent. The subscript $t-2$ denotes variables lagged by two years.

Table 2: Response of Business Revenue to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 18.04*** | 16.95** | 16.95** | 15.88*** | 17.56*** |
| | (5.361) | (6.312) | (6.312) | (4.482) | (4.453) |
| $R^2$ | 0.149 | 0.151 | 0.151 | 0.147 | 0.150 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,468,423 | 2,362,540 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 16.46 | 8.594 | 8.594 | 7.521 | 9.515 |
| | (10.10) | (10.26) | (10.26) | (7.947) | (8.205) |
| $R^2$ | 0.190 | 0.198 | 0.198 | 0.194 | 0.200 |
| Observations | 2,470,015 | 2,470,015 | 2,470,015 | 2,138,690 | 2,038,840 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of firm-level real business revenue to an expansionary monetary policy shock, estimated from the local projection in Equation (2). The dependent variable is the cumulative growth rate of real business revenue, deflated by the GDP deflator, between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 3: Response of Employment to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 5.998*** | 4.500** | 4.500** | 4.209** | 2.683 |
| | (1.418) | (1.663) | (1.663) | (1.632) | (1.641) |
| $R^2$ | 0.143 | 0.145 | 0.145 | 0.137 | 0.142 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,406,130 | 2,301,308 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 6.553* | 2.541 | 2.541 | 1.712 | −0.456 |
| | (3.296) | (3.700) | (3.700) | (2.740) | (2.055) |
| $R^2$ | 0.193 | 0.197 | 0.197 | 0.195 | 0.202 |
| Observations | 2,464,284 | 2,464,284 | 2,464,284 | 2,086,115 | 1,987,350 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of firm-level employment to an expansionary monetary policy shock, estimated from the local projection in Equation (2). The dependent variable is the cumulative growth rate of the number of employees aged 26–65 between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as employment growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4: Response of Average Wage to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 5.870*** | 3.642*** | 3.642*** | 4.386*** | 3.147*** |
| | (1.069) | (1.103) | (1.103) | (1.133) | (0.882) |
| $R^2$ | 0.136 | 0.138 | 0.138 | 0.127 | 0.132 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,413,862 | 2,308,155 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 8.518*** | 5.469** | 5.469** | 5.606*** | 4.057** |
| | (2.070) | (2.228) | (2.228) | (1.865) | (1.649) |
| $R^2$ | 0.169 | 0.172 | 0.172 | 0.169 | 0.175 |
| Observations | 2,470,595 | 2,470,595 | 2,470,595 | 2,097,071 | 1,997,338 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of the firm-level average wage to an expansionary monetary policy shock, estimated from the local projection in Equation (2). The dependent variable is the cumulative growth rate of the real average wage per employee, deflated by the GDP deflator, between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 5: Response of Average Dividend to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 18.02* | 9.582 | 9.582 | 13.50* | 13.49** |
| | (9.383) | (11.15) | (11.15) | (6.622) | (5.046) |
| $R^2$ | 0.130 | 0.157 | 0.157 | 0.121 | 0.125 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,168,282 | 2,075,172 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 36.84*** | 25.09*** | 25.09*** | 28.69*** | 31.51*** |
| | (6.457) | (8.281) | (8.281) | (6.200) | (5.056) |
| $R^2$ | 0.160 | 0.183 | 0.183 | 0.162 | 0.167 |
| Observations | 2,470,337 | 2,470,337 | 2,470,337 | 1,893,886 | 1,805,476 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of the firm-level average dividend to an expansionary monetary policy shock, estimated from the local projection in Equation (2). The dependent variable is the cumulative growth rate of the real average dividend per individual shareholder, deflated by the GDP deflator, between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 6: Response of Business Revenue by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0694^{***}$ | $-0.0428^{***}$ | $-0.0428^{***}$ | $-0.0332^{***}$ | $-0.0327^{***}$ |
| | (0.00812) | (0.00527) | (0.00527) | (0.00567) | (0.00541) |
| $-\hat{MP}_{t-1}$ | $17.84^{***}$ | $16.76^{**}$ | $16.76^{**}$ | $15.75^{***}$ | $17.44^{***}$ |
| | (5.345) | (6.291) | (6.291) | (4.438) | (4.397) |
| Exposure $\times -\hat{MP}_{t-1}$ | 0.0142 | 0.0205 | 0.0205 | 0.0135 | 0.0128 |
| | (0.0163) | (0.0154) | (0.0154) | (0.0157) | (0.0165) |
| $R^2$ | 0.149 | 0.151 | 0.151 | 0.147 | 0.150 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,468,423 | 2,362,540 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.176^{***}$ | $-0.106^{***}$ | $-0.106^{***}$ | $-0.0867^{***}$ | $-0.0811^{***}$ |
| | (0.0170) | (0.00939) | (0.00939) | (0.00937) | (0.00840) |
| $-\hat{MP}_{t-1}$ | 16.10 | 8.433 | 8.433 | 7.440 | 9.443 |
| | (9.988) | (10.14) | (10.14) | (7.804) | (8.044) |
| Exposure $\times -\hat{MP}_{t-1}$ | 0.0221 | 0.0183 | 0.0183 | 0.00863 | 0.00626 |
| | (0.0243) | (0.0233) | (0.0233) | (0.0242) | (0.0256) |
| $R^2$ | 0.191 | 0.199 | 0.199 | 0.195 | 0.200 |
| Observations | 2,470,015 | 2,470,015 | 2,470,015 | 2,138,690 | 2,038,840 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level real business revenue to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3). The dependent variable is the cumulative growth rate of real business revenue, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times -\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 7: Response of Employment by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0595^{***}$ | $-0.0359^{***}$ | $-0.0359^{***}$ | $-0.0255^{***}$ | $-0.0306^{***}$ |
| | (0.00710) | (0.00563) | (0.00563) | (0.00560) | (0.00434) |
| $-\hat{MP}_{t-1}$ | $5.731^{***}$ | $4.266^{**}$ | $4.266^{**}$ | $4.000^{**}$ | 2.463 |
| | (1.344) | (1.603) | (1.603) | (1.576) | (1.522) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0220 | 0.0261 | 0.0261 | 0.0225 | 0.0232 |
| | (0.0170) | (0.0174) | (0.0174) | (0.0167) | (0.0164) |
| $R^2$ | 0.143 | 0.145 | 0.145 | 0.138 | 0.142 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,406,130 | 2,301,308 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.128^{***}$ | $-0.0821^{***}$ | $-0.0821^{***}$ | $-0.0648^{***}$ | $-0.0690^{***}$ |
| | (0.0122) | (0.00779) | (0.00779) | (0.00747) | (0.00620) |
| $-\hat{MP}_{t-1}$ | $6.104^{*}$ | 2.198 | 2.198 | 1.470 | $-0.745$ |
| | (3.207) | (3.642) | (3.642) | (2.708) | (1.902) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0371 | 0.0388 | 0.0388 | 0.0260 | 0.0297 |
| | (0.0242) | (0.0252) | (0.0252) | (0.0256) | (0.0247) |
| $R^2$ | 0.193 | 0.197 | 0.197 | 0.195 | 0.202 |
| Observations | 2,464,284 | 2,464,284 | 2,464,284 | 2,086,115 | 1,987,350 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level employment to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3). The dependent variable is the cumulative growth rate of the number of employees aged 26–65 between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as employment growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 8: Response of Average Wage by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0188^{***}$ | $0.00655^{*}$ | $0.00655^{*}$ | $0.0127^{***}$ | $0.00696^{*}$ |
| | (0.00474) | (0.00351) | (0.00351) | (0.00389) | (0.00329) |
| $-\hat{MP}_{t-1}$ | $5.993^{***}$ | $3.763^{***}$ | $3.763^{***}$ | $4.502^{***}$ | $3.265^{***}$ |
| | (1.079) | (1.105) | (1.105) | (1.168) | (0.895) |
| Exposure $\times -\hat{MP}_{t-1}$ | $-0.0159$ | $-0.0136$ | $-0.0136$ | $-0.0125$ | $-0.0126$ |
| | (0.0120) | (0.0112) | (0.0112) | (0.0112) | (0.0114) |
| $R^2$ | 0.136 | 0.138 | 0.138 | 0.127 | 0.132 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,413,862 | 2,308,155 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.0254^{***}$ | $0.00964^{*}$ | $0.00964^{*}$ | $0.0175^{***}$ | $0.0101^{*}$ |
| | (0.00872) | (0.00534) | (0.00534) | (0.00537) | (0.00509) |
| $-\hat{MP}_{t-1}$ | $8.525^{***}$ | $5.487^{**}$ | $5.487^{**}$ | $5.663^{**}$ | $4.117^{**}$ |
| | (2.228) | (2.362) | (2.362) | (1.958) | (1.744) |
| Exposure $\times -\hat{MP}_{t-1}$ | $-0.00331$ | $-0.00205$ | $-0.00205$ | $-0.00603$ | $-0.00634$ |
| | (0.0250) | (0.0230) | (0.0230) | (0.0230) | (0.0228) |
| $R^2$ | 0.169 | 0.172 | 0.172 | 0.169 | 0.175 |
| Observations | 2,470,595 | 2,470,595 | 2,470,595 | 2,097,071 | 1,997,338 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average wage to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3). The dependent variable is the cumulative growth rate of the real average wage per employee, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times -\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 9: Response of Average Dividend by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.121^{***}$ | 0.0153 | 0.0153 | $0.0593^{***}$ | $0.0599^{***}$ |
| | (0.0185) | (0.0120) | (0.0120) | (0.00867) | (0.0103) |
| $-\hat{MP}_{t-1}$ | $17.40^{*}$ | 8.878 | 8.878 | $12.68^{*}$ | $12.64^{**}$ |
| | (9.424) | (11.10) | (11.10) | (6.533) | (4.945) |
| Exposure $\times -\hat{MP}_{t-1}$ | $0.0526^{*}$ | $0.0801^{***}$ | $0.0801^{***}$ | $0.0886^{***}$ | $0.0930^{***}$ |
| | (0.0296) | (0.0222) | (0.0222) | (0.0225) | (0.0218) |
| $R^2$ | 0.130 | 0.157 | 0.157 | 0.121 | 0.125 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,168,282 | 2,075,172 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.172^{***}$ | 0.0127 | 0.0127 | $0.0705^{***}$ | $0.0719^{***}$ |
| | (0.0289) | (0.0162) | (0.0162) | (0.0128) | (0.0135) |
| $-\hat{MP}_{t-1}$ | $36.58^{***}$ | $24.79^{***}$ | $24.79^{***}$ | $28.28^{***}$ | $31.08^{***}$ |
| | (6.328) | (8.162) | (8.162) | (5.956) | (4.683) |
| Exposure $\times -\hat{MP}_{t-1}$ | 0.0113 | 0.0339 | 0.0339 | 0.0451 | 0.0474 |
| | (0.0762) | (0.0667) | (0.0667) | (0.0662) | (0.0686) |
| $R^2$ | 0.160 | 0.183 | 0.183 | 0.162 | 0.167 |
| Observations | 2,470,337 | 2,470,337 | 2,470,337 | 1,893,886 | 1,805,476 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average dividend to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3). The dependent variable is the cumulative growth rate of the real average dividend per individual shareholder, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times -\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 10: Individual-Level Statistics by Lagged Total Income Rank

| | Percentile of total income at $t-2$ | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 | 50–60 | 60–70 | 70–80 | 80–90 | 90–99 | Top 1% |
| Wage | 260 | 299 | 349 | 405 | 470 | 544 | 644 | 792 | 1,051 | 1,655 | 4,092 |
| | (174) | (205) | (217) | (218) | (236) | (263) | (312) | (394) | (530) | (1,091) | (7,490) |
| Dividend | 42 | 57 | 57 | 59 | 63 | 74 | 94 | 129 | 185 | 455 | 5,783 |
| | (591) | (249) | (209) | (416) | (218) | (487) | (389) | (384) | (572) | (1,361) | (37,061) |
| Total Income | 289 | 342 | 389 | 445 | 510 | 593 | 707 | 883 | 1,184 | 2,023 | 9,448 |
| | (619) | (273) | (246) | (420) | (247) | (474) | (400) | (412) | (599) | (1,469) | (36,587) |
| Dividend Ratio$_{t-2}$ | 0.12 | 0.14 | 0.12 | 0.11 | 0.10 | 0.10 | 0.11 | 0.13 | 0.14 | 0.20 | 0.46 |
| | (0.31) | (0.33) | (0.30) | (0.29) | (0.27) | (0.27) | (0.28) | (0.29) | (0.30) | (0.34) | (0.43) |
| Total Income$_{t-2}$ | 246 | 295 | 352 | 412 | 480 | 560 | 669 | 837 | 1,138 | 2,016 | 9,515 |
| | (32) | (34) | (34) | (35) | (38) | (43) | (56) | (81) | (139) | (604) | (28,764) |
| Age | 43.2 | 43.5 | 42.9 | 42.2 | 42.1 | 42.2 | 42.4 | 43.2 | 45.6 | 48.3 | 51.4 |
| | (9.8) | (10.0) | (9.9) | (9.7) | (9.6) | (9.4) | (9.1) | (8.9) | (8.9) | (8.7) | (8.3) |
| Total observations | 33,167,054 | | | | | | | | | | |

*Notes:* This table reports means and standard deviations (in parentheses) of individual-level variables by groups of lagged total income. The sample consists of individuals registered in Taiwan and aged 26–65 who receive wage or dividend income from firms in our final firm-level sample over 2004–2023. Individuals are sorted each year into eleven groups according to their total income at $t-2$, defined as the sum of wage and dividend income; the top decile is further split into the 90th–99th percentiles and the top 1%. Wage, Dividend, and Total Income are measured in year $t$. Wage consists of salaries and bonuses from employers. Dividend Ratio$_{t-2}$ is the ratio of dividend income to total income at $t-2$. All monetary variables are in thousands of NT dollars and deflated by the GDP deflator. Observations with absolute annual growth exceeding 500% are excluded; since this restriction is applied separately to each income component, the means of wage and dividend income need not sum to the mean of total income. Observations are individual-year pairs.

# Appendix A Alternative Monetary Policy

In this section, we check whether our results are robust to an alternative identification approach following Bauer and Swanson (2023). Their method has two steps. First, they measure the monetary policy surprise as the first principal component of intraday changes in the first four quarterly Eurodollar futures contracts (ED1–ED4) around FOMC announcements. Second, they remove the predictable component of this surprise by regressing it on macroeconomic and financial variables observed before each meeting and keeping the residual. Their predictors are the nonfarm payrolls surprise, year-over-year payroll employment growth, the three-month log change in the S&P 500, the three-month change in the slope of the Treasury yield curve, the three-month log change in commodity prices, and the options-implied skewness of the 10-year Treasury yield. Although their implementation uses intraday data, the same approach can be applied to daily data.

Because intraday data are not available for Taiwanese interest rates, we measure the monetary policy surprise as the first principal component of the daily changes in the 30-, 90-, and 180-day repurchase agreement rates from the day before to the day of each policy meeting of the Central Bank of Taiwan. Following the spirit of Bauer and Swanson (2023), we select predictors that are publicly observed before each meeting and plausibly enter the central bank's reaction function, adapting their variable list to the data available for Taiwan. Specifically, we regress this surprise on the following variables, all observed before the meeting: (i) year-over-year growth in Taiwanese payroll employment in the month before the meeting; (ii) year-over-year inflation in the Taiwanese price index of daily necessities in the month before the meeting; (iii) the log change in the TAIEX from 85 days before to one day before the meeting; (iv) the change in the slope of the Taiwanese government bond yield curve over the same period, measured as the second principal component of the 2-, 5-, 10-, and 20-year yields; (v) year-over-year growth in U.S. nonfarm payroll employment in the month before the meeting; (vi) the log change in the S&P 500 from 85 days before to one day before the meeting; and (vii) the change in the slope of the U.S. Treasury yield curve over the same period, measured as the second principal component of the 1-, 2-, 3-, 5-, 7-, 10-, 20-, and 30-year yields. All year-over-year growth rates are computed as log differences,

and all windows are measured in calendar days. We omit the nonfarm payrolls surprise and the Treasury yield skewness used by Bauer and Swanson (2023), because survey forecasts of Taiwanese employment releases and options on Taiwanese government bonds are not available. The residual from this regression is our orthogonalised monetary policy surprise. Because a principal component has no natural scale, we normalise this surprise in the same way as the baseline shock, so that an expansionary shock lowers the overnight interbank rate by 12.5 basis points. The estimated responses under the two shocks are therefore comparable in terms of their impact on the overnight interbank rate. Note, however, that the two series extract policy surprises from different segments of the yield curve: the baseline shock uses the two-year government bond yield as its benchmark and incorporates information from longer maturities, whereas the alternative shock is based on short-term repurchase agreement rates. Differences in the magnitude of the estimated responses may therefore partly reflect the different information captured by the two series. Taiwanese data are from the Taiwan Economic Journal (TEJ), and U.S. data are from FRED.

## Appendix A.1 Aggregate Responses

We first re-estimate the aggregate impulse responses in Figure 2, replacing the baseline shock with the orthogonalised monetary policy surprise. As in the baseline, the shock is normalised to lower the overnight interbank rate by 12.5 basis points, and the shaded regions denote 90% confidence intervals constructed using Newey-West standard errors. Figure A1 reports the results.

The overnight interbank rate falls by about 0.125 percentage points on impact and continues to decline to roughly 0.3 percentage points by month 4. The decline is statistically significant during the first five months, after which the rate gradually returns toward zero. The point estimate falls again after month 16, but this later decline is imprecisely estimated. As in the baseline, the shock therefore captures a short-lived easing of monetary conditions in Taiwan.

The responses of real activity and prices are similar to the baseline. Industrial production rises significantly between months 5 and 8, peaking at about 10 log points around month 8, compared with a significant increase between months 5 and 9 in the baseline. The CPI

rises gradually and peaks at about 1.2 log points around month 13, similar in magnitude to the baseline increase of about 1 log point after 12 months, although the lower bound of the confidence interval stays close to zero.

The trade responses also follow the baseline pattern, although they occur slightly later. Imports rise significantly between about months 11 and 14, peaking at about 15 log points, and exports rise significantly between about months 10 and 14, peaking at about 11 log points. In the baseline, both variables rise significantly between months 6 and 14, by about 15 log points each. The real effective exchange rate begins to depreciate after month 12, and the depreciation becomes significant around months 18 to 22, reaching about 5 log points, compared with a similar depreciation after month 12 in the baseline. Beyond month 16, the point estimates for industrial production, imports, and exports turn negative, but the confidence intervals widen considerably and none of these responses is statistically significant.

Overall, the orthogonalised shock reproduces the main aggregate responses of the baseline: an expansionary shock is followed by a temporary decline in the overnight interbank rate, a rise in industrial production, a gradual increase in consumer prices, an expansion of both imports and exports, and a subsequent real depreciation. The magnitudes are comparable to the baseline, while the responses of trade and the exchange rate occur somewhat later.

## Appendix A.2 Firms' Response to Monetary Policy

Tables A1–A4 re-estimate Equation (2), replacing our baseline shock with the orthogonalised monetary policy surprise described above. The outcome variables and specifications follow those in Tables 2–5. Each table reports the estimates for $h = 0$ in Panel A and for $h = 1$ in Panel B, columns (1) through (5) add the same sets of controls in the same order, and standard errors are two-way clustered at the firm and year levels.

Table A1 shows that the revenue response retains the pattern documented in our baseline results. In Panel A, revenue rises by between 8.018% and 10.05% at $h = 0$, and the estimates are statistically significant in all five specifications. In Panel B, the cumulative response at $h = 1$ remains positive but is statistically insignificant in every specification. In our preferred specification, column (5), revenue increases by 9.749% at $h = 0$ and by a cumulative 5.095%

at $h = 1$. These estimates are roughly half the size of their baseline counterparts (17.56% and 9.515%), but the response is again strongest on impact and diminishes by $h = 1$.

Table A2 reports the employment response. At $h = 0$, employment rises by between 2.808% and 3.766%, within the range of the baseline estimates. Unlike in the baseline, however, the estimate is statistically significant in all five specifications, including column (5), where employment increases by 2.856%. At $h = 1$, the cumulative response shrinks towards zero and is statistically insignificant in every specification; the estimate in column (5) is 0.226%. The alternative shock therefore provides somewhat stronger evidence of a short-lived increase in employment, but it confirms that the employment response does not persist and remains modest relative to the responses of revenue and dividends.

Table A3 shows that the wage response is closely aligned with our baseline. Average wages rise by between 2.429% and 3.647% at $h = 0$, with all five estimates statistically significant. The cumulative increase at $h = 1$ ranges from 2.993% to 4.134% and remains statistically significant in all five specifications, albeit only at the 10% level in columns (2) through (4). In column (5), average wages rise by 3.647% at $h = 0$ and by a cumulative 4.134% at $h = 1$, close to the baseline estimates of 3.147% and 4.057%. As in the baseline, the wage response shows no sign of fading by $h = 1$.

Table A4 reports the response of average dividends per shareholder. At $h = 0$, the point estimates are positive in all five specifications, ranging from 2.679% to 8.277%, but none is statistically significant. By $h = 1$, the cumulative increase grows to between 9.560% and 19.96% and is statistically significant in columns (1), (4), and (5). In our preferred specification, column (5), average dividends rise by 8.277% at $h = 0$ and by a cumulative 19.96% at $h = 1$, the latter significant at the 1% level. Although these estimates are smaller than their baseline counterparts (13.49% and 31.51%), the dividend response again builds up over time and, at $h = 1$, is nearly five times as large as the corresponding wage response, compared with nearly eight times in the baseline.

Overall, the results based on the alternative shock support our main conclusions. An expansionary monetary policy shock raises firm revenue on impact, with the effect diminishing by $h = 1$; it raises average wages persistently; and it raises average dividends by considerably more than average wages at $h = 1$. The employment response remains modest

and short-lived. The differences from the baseline lie mainly in magnitude: the revenue and dividend responses are smaller, while the employment response at $h = 0$ becomes statistically significant. These differences may partly reflect the different construction of the two shock series, as the alternative measure removes the component of policy surprises that is predictable from information available before each meeting. Importantly, however, they do not alter the qualitative pattern of our findings, in particular the stronger response of dividends than wages.

## Appendix A.3 Credit Channel and Dividend Payout

Tables A5–A8 re-estimate Equation (3), replacing our baseline shock with the orthogonalised monetary policy surprise. The exposure measure, specifications, and clustering are the same as in Tables 6–9. Unlike a change in the price deflator, a change in the shock series alters the variation used to identify the interaction coefficient $\gamma^h$, and therefore provides an independent check on the cross-sectional pattern documented in our baseline results.

Table A5 reports the results for business revenue. The interaction estimates are small and statistically insignificant in all five specifications at both horizons. Although their signs are mostly negative, in contrast to the positive baseline estimates, neither set of estimates is statistically distinguishable from zero. In our preferred specification, column (5), a one-percentage-point higher short-term loan ratio is associated with a 0.00574-percentage-point smaller revenue response at $h = 0$ and an additional 0.00192-percentage-point cumulative response at $h = 1$. As in the baseline, we find no evidence that revenue responds differently across firms with different short-term loan ratios.

Table A6 reports the corresponding results for employment. The interaction estimates are close to zero and statistically insignificant in all five specifications at both horizons. In column (5), a one-percentage-point higher short-term loan ratio is associated with an additional 0.00263-percentage-point employment response at $h = 0$ and an additional 0.00390-percentage-point cumulative response at $h = 1$, smaller than the baseline estimates of 0.0232 and 0.0297 percentage points. We therefore again find no statistically significant difference in employment responses by short-term loan exposure.

Table A7 turns to average wages per employee. As in the baseline, the interaction es-

timates are negative at both horizons, but none is statistically significant. In column (5), a one-percentage-point higher short-term loan ratio is associated with a 0.0120-percentage-point smaller wage response at $h = 0$ and a 0.0135-percentage-point smaller cumulative response at $h = 1$, compared with 0.0126 and 0.00634 percentage points in the baseline. These estimates provide no evidence that firms with greater short-term borrowing raise average wages more strongly following the policy shock.

Table A8 reports the results for average dividends per shareholder. In Panel A, the interaction estimates are positive and statistically significant across all five specifications, at the 5% level in columns (2), (3), and (5) and at the 10% level in columns (1) and (4). In column (5), a one-percentage-point higher short-term loan ratio increases the estimated dividend response at $h = 0$ by 0.0565 percentage points, compared with 0.0930 percentage points in the baseline. In Panel B, the interaction estimates are positive but close to zero and statistically insignificant; the column (5) estimate is 0.00929 percentage points. Firms with greater short-term borrowing therefore again exhibit a stronger initial dividend response that does not persist at $h = 1$.

Overall, the alternative shock reproduces the cross-sectional pattern of our baseline results. The only significant interaction is again for dividends at $h = 0$, while the interactions for revenue, employment, and wages remain statistically insignificant. The differential dividend response is smaller than in the baseline, at roughly 60% of the baseline estimate in our preferred specification, consistent with the smaller average dividend response under this shock documented above. Our evidence that short-term borrowing exposure amplifies the initial dividend response to expansionary monetary policy is therefore robust to the identification of monetary policy shocks.

## Appendix A.4 Distributional Implications for Individual Income

We next re-estimate the individual-level specification in equation (4), replacing the baseline shock with the orthogonalised monetary policy surprise constructed above. The sample, income-group assignment, fixed effects, and clustering are the same as in the baseline. Figure A2 reports the results, using the same layout as Figure 3. Although both shocks are normalised to the same impact on the overnight interbank rate, they capture policy sur-

prises from different segments of the yield curve. We therefore focus primarily on the shape of the responses across the income distribution, while also reporting the baseline estimates for comparison.

The wage contribution displays the same pattern as in the baseline. At $h = 0$, it rises with pre-shock income rank, from 2.369 percentage points in the bottom decile to 6.554 percentage points in the 90th–99th percentile group, and is lower, at 5.552 percentage points, for the top 1%. The baseline estimates for the same groups are 4.227, 9.321, and 7.098 percentage points. At $h = 1$, the estimates are around three percentage points for most of the distribution and are not statistically significant, so, as in the baseline, the income gradient in wages does not persist.

The dividend contribution again responds most strongly at the very top. At $h = 0$, it is 8.200 percentage points for the top 1%, compared with 2.445 percentage points for the bottom decile and 2.733 percentage points for the 90th–99th percentile group (11.13, 4.152, and 3.845 percentage points in the baseline). Dividend contributions for the other income groups are positive but less precisely estimated than in the baseline. At $h = 1$, the estimate for the top 1% is 7.736 percentage points but, unlike the baseline estimate of 13.06 percentage points, is not statistically significant.

Adding the two contributions gives the implied total response.[12] At $h = 0$, it rises from about 4.81 percentage points in the bottom decile to 9.29 percentage points in the 90th–99th percentile group and 13.75 percentage points for the top 1%. As in the baseline, the increase from the 90th–99th percentile group to the top 1% (4.47 percentage points) is about as large as the entire increase from the bottom decile to the 90th–99th percentile group (4.47 percentage points), and this jump is driven by dividends, since the top 1% has a smaller wage contribution than the 90th–99th percentile group.

Overall, the alternative shock reproduces the main distributional patterns of the baseline in the year of the shock: the wage contribution increases with income rank up to the 90th–99th percentile, the dividend contribution is distinctly large for the top 1%, and the implied total income gain peaks at the very top. The main difference is that the persistence of the

[12] As in the baseline, the implied totals are sums of point estimates from separate regressions and are therefore approximate.

top-income response at $h = 1$ is weaker and imprecisely estimated under the alternative shock. Our conclusion that expansionary monetary policy tilts contemporaneous income gains toward top earners, largely through dividends, is therefore not driven by the particular choice of monetary policy shock.

# Appendix B Alternative Deflators

In our baseline analysis, all nominal variables are deflated by the GDP deflator. In this section, we instead deflate them by the consumer price index (CPI). Whereas the GDP deflator reflects the prices of goods and services produced in the economy, the CPI reflects the prices faced by households as consumers, and therefore measures changes in the purchasing power of wage and dividend income. All other aspects of the sample construction and empirical specification are unchanged. Because employment is measured as a headcount and does not depend on the choice of deflator, we focus on business revenue, average wages, and average dividends.[13]

## Appendix B.1 Firms' Response to Monetary Policy

Tables A9–A11 report the results. As in the baseline, each table reports the estimates for $h = 0$ in Panel A and for $h = 1$ in Panel B, columns (1) through (5) add the same sets of controls in the same order, and standard errors are two-way clustered at the firm and year levels.

Table A9 shows that the revenue response remains similar to our baseline. In Panel A, revenue rises by between 12.45% and 14.74% at $h = 0$, and the estimates are statistically significant in all five specifications. In Panel B, the cumulative response at $h = 1$ remains positive but is statistically insignificant in every specification. In our preferred specification, column (5), revenue increases by 14.13% at $h = 0$ and by a cumulative 3.791% at $h = 1$, compared with 17.56% and 9.515% in the baseline. The response is therefore again strongest on impact and diminishes by $h = 1$.

Table A10 shows that, unlike our baseline, the average wage response is no longer statistically significant once wages are deflated by the CPI. At $h = 0$, the estimate is positive and significant only in column (1), and it declines towards zero as controls are added. At $h = 1$, none of the five estimates is statistically significant. In column (5), the estimated wage response is $-0.167\%$ at $h = 0$ and $-1.258\%$ at $h = 1$, neither of which is statistically

[13] The employment estimates differ only marginally from the baseline, because the trimming of extreme growth rates is applied to the deflated variables and therefore slightly alters the estimation sample.

distinguishable from zero.

Table A11 reports the response of average dividends per shareholder. At $h = 0$, the point estimates range from 6.777% to 14.95%, but only the estimate in column (5) is statistically significant, at the 10% level. By $h = 1$, the cumulative increase ranges from 19.65% to 30.12% and is statistically significant in all five specifications. In our preferred specification, average dividends rise by 10.10% at $h = 0$ and by a cumulative 25.64% at $h = 1$, compared with 13.49% and 31.51% in the baseline.

These differences from the baseline follow a systematic pattern. Because the deflator is common to all firms in a given year, replacing the GDP deflator with the CPI shifts the real growth rate of every outcome by approximately the same amount, namely the difference between CPI inflation and GDP deflator inflation over the relevant horizon. Consistent with this, relative to the baseline, the column (5) estimates for revenue, wages, and dividends decline by 3.4, 3.3, and 3.4 percentage points, respectively, at $h = 0$, and by 5.7, 5.3, and 5.9 percentage points at $h = 1$. This suggests that, in our sample period, consumer prices rose by more than the GDP deflator following an expansionary shock. Crucially, the gap between the dividend and wage responses is virtually unaffected: in column (5), dividends exceed wages by 10.27 percentage points at $h = 0$ and 26.90 percentage points at $h = 1$, compared with 10.34 and 27.45 percentage points in the baseline.

The two sets of wage estimates therefore carry different economic interpretations. Deflated by the GDP deflator, the wage response measures wages relative to the prices of output, which is relevant to firms' labour costs. Deflated by the CPI, it measures workers' purchasing power. Taken together, our results indicate that firms raise wages relative to output prices following an expansionary shock, but that nominal wage growth roughly keeps pace with consumer prices, leaving workers' real purchasing power broadly unchanged. By contrast, dividends rise substantially under either deflator. Our central finding, that dividend income responds considerably more strongly than wage income to an expansionary monetary policy shock, is therefore robust to the choice of deflator.

## Appendix B.2 Credit Channel and Dividend Payout

Tables A12–A14 re-estimate Equation (3) using CPI-deflated outcomes. The exposure measure, specifications, and clustering are the same as in Tables 6–9. As before, we do not report employment, which does not depend on the choice of deflator. Because the deflator is common to all firms in a given year, replacing the GDP deflator with the CPI mainly affects the coefficient on the monetary policy shock, $\beta_1^h$, in the same way as documented above. The interaction coefficient $\gamma^h$, by contrast, is identified from differences in responses across firms with different short-term loan ratios, and should therefore be largely unaffected.

Table A12 reports the results for business revenue. In both panels, the interaction estimates are positive but statistically insignificant in all five specifications. In our preferred specification, column (5), a one-percentage-point higher short-term loan ratio is associated with an additional 0.0118-percentage-point revenue response at $h = 0$ and an additional 0.00614-percentage-point cumulative response at $h = 1$, nearly identical to the baseline estimates of 0.0128 and 0.00626 percentage points.

Table A13 turns to average wages per employee. As in the baseline, the interaction estimates are negative at both horizons, but none is statistically significant. In column (5), a one-percentage-point higher short-term loan ratio is associated with a 0.0140-percentage-point smaller wage response at $h = 0$ and a 0.00810-percentage-point smaller cumulative response at $h = 1$, compared with 0.0126 and 0.00634 percentage points in the baseline. These estimates again provide no evidence that firms with greater short-term borrowing raise average wages more strongly following the policy shock.

Table A14 reports the results for average dividends per shareholder. In Panel A, the interaction estimates are positive and statistically significant across all five specifications, at the 1% level in columns (2) through (5) and at the 10% level in column (1). In column (5), a one-percentage-point higher short-term loan ratio increases the estimated dividend response at $h = 0$ by 0.0922 percentage points, compared with 0.0930 percentage points in the baseline. In Panel B, the interaction estimates remain positive but are statistically insignificant; the column (5) estimate is 0.0422 percentage points, compared with 0.0474 in the baseline. Firms with greater short-term borrowing therefore again exhibit a stronger

initial dividend response, while the differential response does not persist at $h = 1$.

Overall, the cross-sectional pattern is virtually unchanged when outcomes are deflated by the CPI. The only significant interaction is for dividends at $h = 0$, and its magnitude closely matches the baseline estimate. Our evidence that dividend payouts respond more strongly at firms with greater short-term borrowing exposure is therefore robust to the choice of deflator.

## Appendix B.3 Distributional Implications for Individual Income

We next re-estimate the individual-level specification in equation (4), deflating wage and dividend income by the CPI instead of the GDP deflator. The monetary policy shock, income-group assignment, fixed effects, and clustering are the same as in the baseline. Because the 500% absolute-change restriction is applied to the CPI-deflated series, the estimation samples differ slightly from those in the baseline. Figure A3 reports the results, using the same layout as Figure 3. Since the shock is unchanged, the estimates can be compared with the baseline in magnitude as well as in shape.

The wage contribution displays the same gradient as in the baseline but is smaller in magnitude. At $h = 0$, it rises steadily with pre-shock income rank, from 2.169 percentage points in the bottom decile to 7.066 percentage points in the 90th–99th percentile group, and is lower, at 4.570 percentage points, for the top 1%. The baseline estimates for the same groups are 4.227, 9.321, and 7.098 percentage points. The estimates for all groups below the top 1% are significant at the 5% level or better, whereas the estimate for the top 1% is not statistically significant. At $h = 1$, the estimates lie between 1.390 and 2.402 percentage points and none is statistically significant. As in the baseline, the income gradient in wages does not persist beyond the year of the shock, and the wage response in the following year is weaker than in the baseline, where it was significant for the bottom nine deciles.

The dividend contribution is almost identical to the baseline. At $h = 0$, it is 3.865 and 4.398 percentage points for the bottom two deciles, declines gradually to 1.910 percentage points for the 80th–90th percentile, rises to 3.520 percentage points for the 90th–99th percentile group, and jumps to 11.20 percentage points for the top 1% (4.152, 4.685, 2.280, 3.845, and 11.13 percentage points in the baseline). All estimates are significant at the 10% level or better. The top 1% contribution is about 2.9 times that of the bottom decile and 3.2 times that of the 90th–99th percentile group. At $h = 1$, the contributions are larger than at $h = 0$ for every group. They are significant at the 5% level or better for all groups below the top 1%, and the top 1% estimate of 11.65 percentage points is significant at the 10% level, compared with 13.06 percentage points in the baseline.

Adding the two contributions gives the implied total response.[14] At $h = 0$, it is 6.03

[14] As in the baseline, the implied totals are sums of point estimates from separate regressions and are

percentage points in the bottom decile, remains between 7.15 and 7.62 percentage points for for the 10th–70th percentiles, and rises to 8.45 and 8.49 percentage points for the 70th–80th and 80th–90th percentile, 10.59 percentage points for the 90th–99th percentile, and 15.77 percentage points for the top 1%. The response of the top 1% is thus about 2.6 times that of the bottom decile. As in the baseline, the increase from the 90th–99th percentile group to the top 1% (5.18 percentage points) exceeds the entire increase from the bottom decile to the 90th–99th percentile group (4.55 percentage points). This jump is again driven by dividends, which account for about 71% of the top 1%'s implied total response, since the top 1% has a smaller wage contribution than the 90th–99th percentile group. At $h = 1$, the implied total lies between 4.64 and 7.01 percentage points for all groups below the top 1%, but reaches 13.04 percentage points for the top 1%, so the concentration of gains at the top persists into the following year.

Overall, deflating by the CPI reproduces the main distributional patterns of the baseline. The wage contribution increases with income rank up to the 90th–99th percentile, the dividend contribution is distinctly large and persistent for the top 1%, and the implied total income gain peaks at the very top in both years. The main difference is that the wage contribution is smaller throughout the distribution and is not statistically significant at $h = 1$. Because the dividend responses are virtually unchanged, our conclusion that expansionary monetary policy tilts income gains toward top earners, largely through dividends, is not driven by the choice of price deflator.

therefore approximate.

# Appendix C Alternative Measure of Credit Exposure

In our baseline analysis, we measure firms' exposure to external credit by the ratio of short-term loans to total assets at $t-2$. In this section, we instead use the leverage ratio, defined as total liabilities divided by total assets, also measured at $t-2$. The leverage ratio provides a broader measure of firms' reliance on external financing. It includes long-term debt and non-bank liabilities such as trade payables, which are less directly affected by changes in short-term policy rates, and it is more evenly distributed across firms than the short-term loan ratio, whose median is zero (Table 1). All other aspects of the specification are unchanged.

Tables A15–A18 report the results. Table A15 shows that the revenue interaction estimates are statistically insignificant in all five specifications at both horizons. In our preferred specification, column (5), a one-percentage-point higher leverage ratio is associated with an additional 0.0113-percentage-point revenue response at $h = 0$ and a 0.00405-percentage-point smaller cumulative response at $h = 1$. As in the baseline, we find no evidence that revenue responds differently across firms with different exposure to external credit.

Table A16 reports the corresponding results for employment. The interaction estimates are positive but statistically insignificant in all five specifications at both horizons. In column (5), a one-percentage-point higher leverage ratio is associated with an additional 0.0184-percentage-point employment response at $h = 0$ and an additional 0.0226-percentage-point cumulative response at $h = 1$, close to the baseline estimates of 0.0232 and 0.0297 percentage points.

Table A17 turns to average wages per employee. At $h = 0$, the interaction estimates are negative in all five specifications, and at $h = 1$ they are close to zero with mixed signs; none is statistically significant. In column (5), a one-percentage-point higher leverage ratio is associated with a 0.0181-percentage-point smaller wage response at $h = 0$ and a 0.00524-percentage-point smaller cumulative response at $h = 1$. These estimates again provide no evidence that firms with greater exposure to external credit raise average wages more strongly following the policy shock.

Table A18 reports the results for average dividends per shareholder. In Panel A, the interaction estimates are positive and statistically significant across all five specifications, at

the 5% level in columns (1), (4), and (5) and at the 10% level in columns (2) and (3). In column (5), a one-percentage-point higher leverage ratio increases the estimated dividend response at $h = 0$ by 0.0734 percentage points, compared with 0.0930 percentage points for the short-term loan ratio in the baseline. In Panel B, the interaction estimates remain positive but are statistically insignificant. Although the column (5) point estimate at $h = 1$ (0.0735 percentage points) is similar in size to that at $h = 0$, it is imprecisely estimated, and we therefore find no statistically significant evidence of a differential response at this horizon.

Overall, the results based on the leverage ratio closely mirror our baseline findings. The only significant interaction is for dividends at $h = 0$, while the interactions for revenue, employment, and wages remain statistically insignificant. The differential dividend response per percentage point of exposure is somewhat smaller than in the baseline, which is consistent with the leverage ratio including liabilities that are less sensitive to short-term policy rates. Our evidence that firms more exposed to external credit raise dividends more strongly following an expansionary monetary policy shock is therefore robust to the measure of credit exposure.

# Figure for Appendix

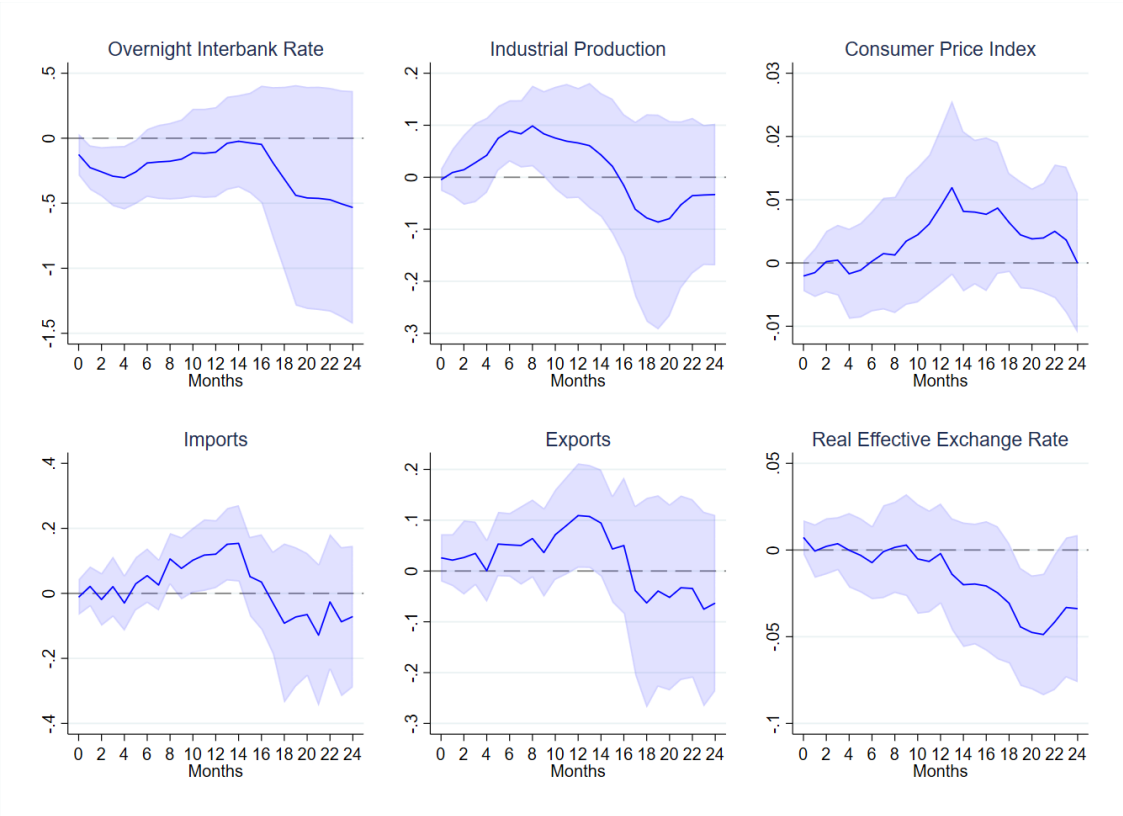
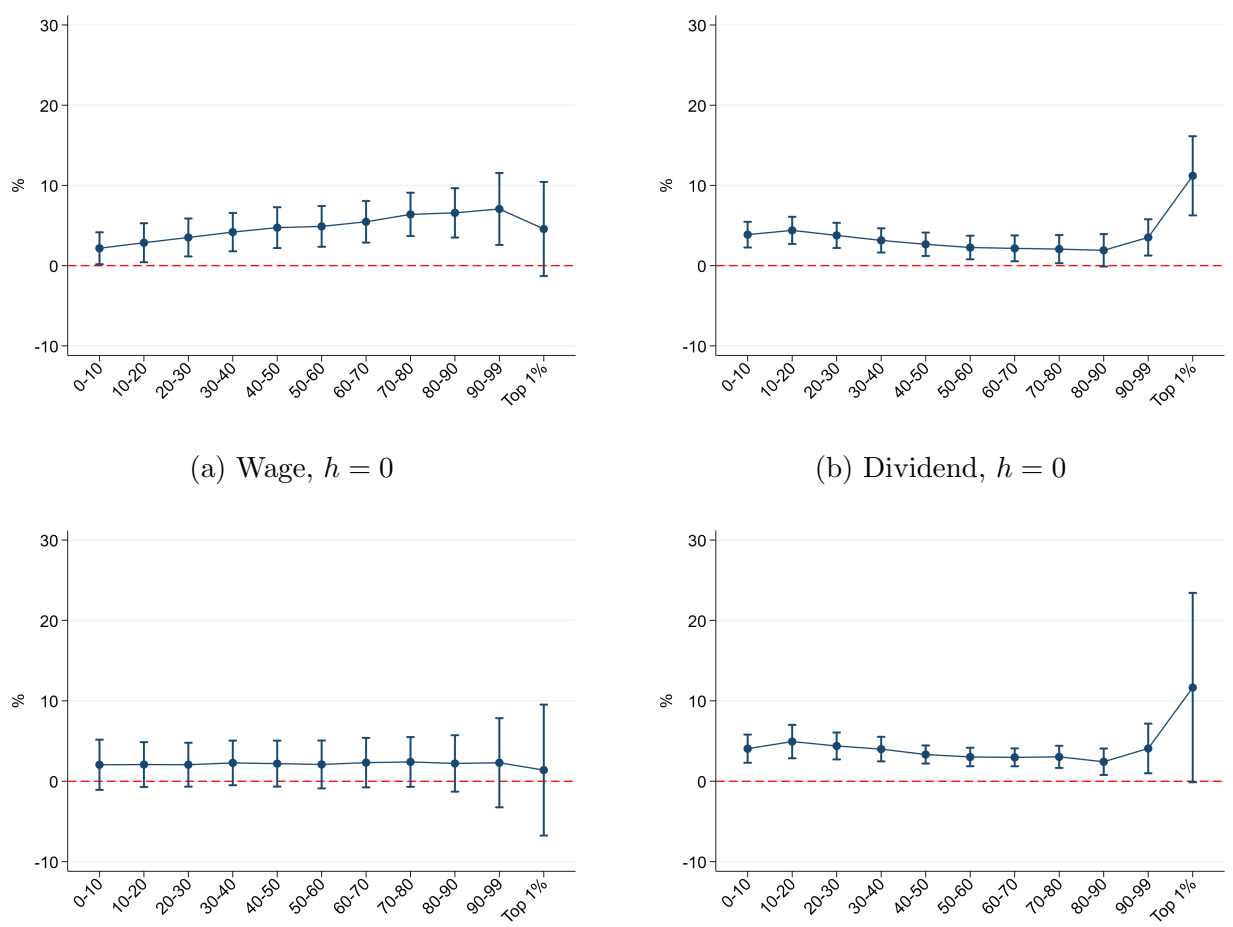



Figure A1: Impulse Responses to Alternative Shocks

*Note:* This figure shows the cumulative impulse responses of six Taiwanese macroeconomic variables to an expansionary monetary policy shock, using the alternative shock constructed following Bauer and Swanson (2023) as described in Appendix A. The shock is the residual from regressing the first principal component of daily changes in the 30-, 90-, and 180-day repurchase agreement rates around each policy meeting on macroeconomic and financial variables observed before the meeting, and is normalised to lower the overnight interbank rate by 12.5 basis points in the first month. Responses are estimated from the local projection in Equation (1) using monthly data from January 2004 to December 2023, with 12 lags of the outcome variable and 12 lags of the monetary policy shock as controls. The outcome variables and units are the same as in Figure 2. The horizontal axis shows the number of months after the shock. Solid blue lines denote point estimates, and shaded areas denote 90% confidence intervals based on Newey–West standard errors.




(a) Wage, $h = 0$

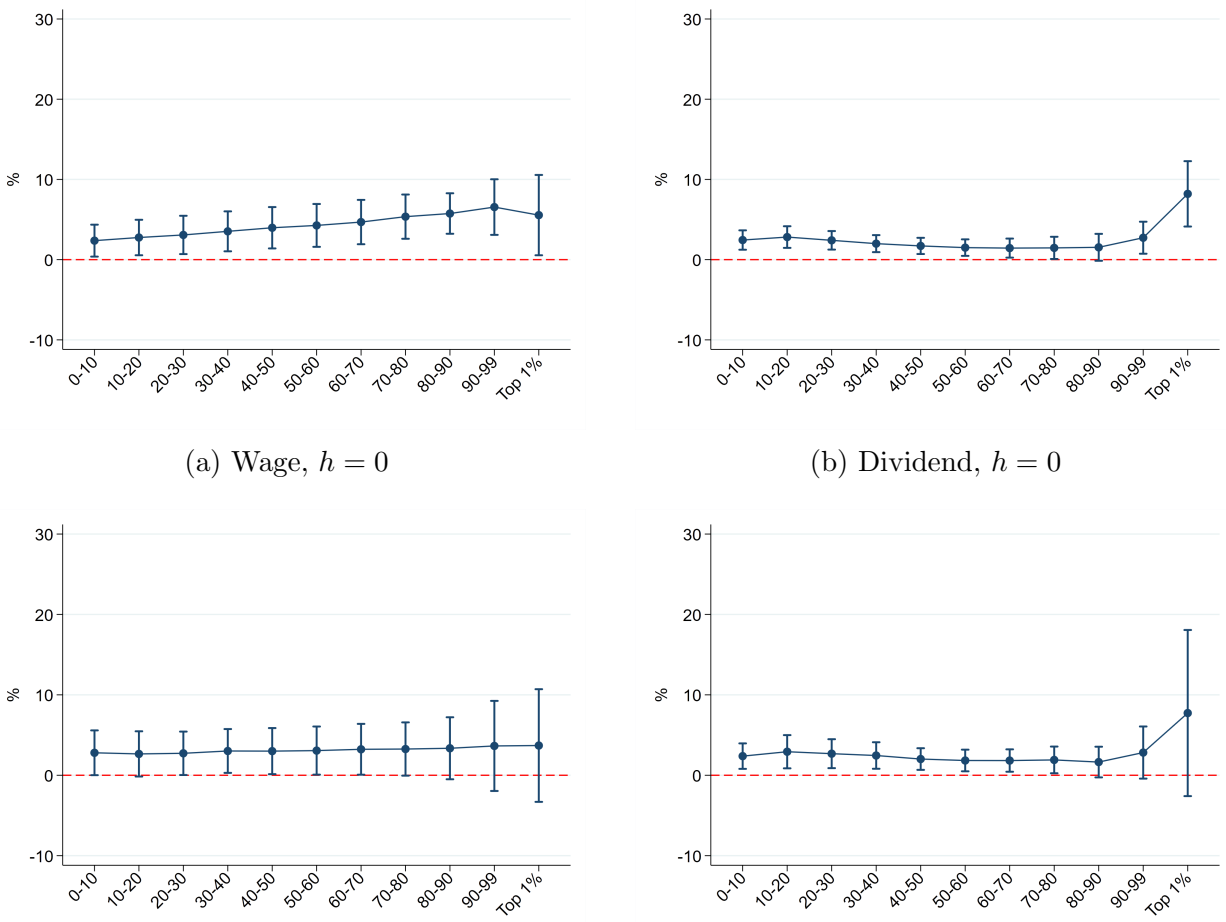
(b) Dividend, $h = 0$

(c) Wage, $h = 1$

(d) Dividend, $h = 1$

Figure A3: Individual Impulse Responses of CPI-deflated Wage and Dividend

*Note:* This figure re-estimates Figure 3 with wage and dividend income deflated by the consumer price index (CPI) instead of the GDP deflator. It shows the estimated contributions of wage income (panels (a) and (c)) and dividend income (panels (b) and (d)) to individual total income growth following an expansionary monetary policy shock that lowers the overnight interbank rate by 12.5 basis points, at $h = 0$ (panels (a) and (b)) and $h = 1$ (panels (c) and (d)), estimated from Equation (4). Individuals are grouped by their total income rank at $t-2$ into the bottom nine deciles, the 90th–99th percentiles, and the top 1%. The monetary policy shock, fixed effects, and income groups are the same as in Figure 3. Because observations with absolute changes exceeding 500% are excluded based on CPI-deflated values, the estimation sample differs slightly from the baseline. Dots denote point estimates, and bars denote 95% confidence intervals based on standard errors two-way clustered at the individual and year levels.




(a) Wage, $h = 0$

(b) Dividend, $h = 0$

(c) Wage, $h = 1$

(d) Dividend, $h = 1$

Figure A2: Individual Impulse Responses of Wage and Dividend to Alternative Shock

*Note:* This figure re-estimates Figure 3 using the alternative shock constructed following Bauer and Swanson (2023) as described in Appendix A, normalised to lower the overnight interbank rate by 12.5 basis points. It shows the estimated contributions of wage income (panels (a) and (c)) and dividend income (panels (b) and (d)) to individual total income growth at $h = 0$ (panels (a) and (b)) and $h = 1$ (panels (c) and (d)), estimated from Equation (4). The dependent variable is the change in each income component between $t-1$ and $t+h$, divided by the individual's total income at $t-1$, so the estimates are expressed in percentage points of total income growth. Individuals are grouped by their total income rank at $t-2$ into the bottom nine deciles, the 90th–99th percentiles, and the top 1%. The sample, fixed effects, and deflator are the same as in Figure 3. Dots denote point estimates, and bars denote 95% confidence intervals based on standard errors two-way clustered at the individual and year levels.

# Table for Appendix

Table A1: Response of Business Revenue to the Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 10.05* | 8.787* | 8.787* | 8.018* | 9.749** |
| | (4.803) | (4.678) | (4.678) | (4.164) | (4.212) |
| $R^2$ | 0.142 | 0.147 | 0.147 | 0.143 | 0.147 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,468,423 | 2,362,540 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 6.735 | 3.986 | 3.986 | 3.692 | 5.095 |
| | (8.776) | (7.789) | (7.789) | (6.436) | (6.029) |
| $R^2$ | 0.186 | 0.198 | 0.198 | 0.192 | 0.198 |
| Observations | 2,470,015 | 2,470,015 | 2,470,015 | 2,138,690 | 2,038,840 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of firm-level real business revenue to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of real business revenue, deflated by the GDP deflator, between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 2. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A2: Response of Employment to the Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 3.766** | 3.138** | 3.138** | 2.808* | 2.856** |
| | (1.578) | (1.408) | (1.408) | (1.534) | (1.291) |
| $R^2$ | 0.142 | 0.145 | 0.145 | 0.138 | 0.142 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,406,130 | 2,301,308 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 2.963 | 1.691 | 1.691 | 0.740 | 0.226 |
| | (2.822) | (2.256) | (2.256) | (1.937) | (1.778) |
| $R^2$ | 0.192 | 0.197 | 0.197 | 0.194 | 0.201 |
| Observations | 2,464,284 | 2,464,284 | 2,464,284 | 2,086,115 | 1,987,350 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of firm-level employment to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of the number of employees aged 26–65 between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as employment growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 3. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A3: Response of Average Wage to the Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 3.229** | 2.429** | 2.429** | 3.274** | 3.647*** |
| | (1.312) | (1.151) | (1.151) | (1.289) | (0.849) |
| $R^2$ | 0.135 | 0.138 | 0.138 | 0.127 | 0.132 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,413,862 | 2,308,155 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 4.114** | 2.993* | 2.993* | 3.461* | 4.134** |
| | (1.882) | (1.557) | (1.557) | (1.786) | (1.428) |
| $R^2$ | 0.168 | 0.172 | 0.172 | 0.168 | 0.175 |
| Observations | 2,470,595 | 2,470,595 | 2,470,595 | 2,097,071 | 1,997,338 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of the firm-level average wage to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of the real average wage per employee, deflated by the GDP deflator, between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A4: Response of Average Dividend to the Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 5.492 | 2.679 | 2.679 | 5.739 | 8.277 |
| | (7.154) | (6.917) | (6.917) | (6.222) | (5.114) |
| $R^2$ | 0.125 | 0.156 | 0.156 | 0.117 | 0.121 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,168,282 | 2,075,172 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 14.02* | 9.560 | 9.560 | 13.99* | 19.96*** |
| | (7.817) | (7.048) | (7.048) | (7.354) | (5.515) |
| $R^2$ | 0.152 | 0.181 | 0.181 | 0.158 | 0.165 |
| Observations | 2,470,337 | 2,470,337 | 2,470,337 | 1,893,886 | 1,805,476 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of the firm-level average dividend to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of the real average dividend per individual shareholder, deflated by the GDP deflator, between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 5. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A5: Response of Business Revenue by Short-Term Loan Ratio, Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0863^{***}$ | $-0.0450^{***}$ | $-0.0450^{***}$ | $-0.0362^{***}$ | $-0.0345^{***}$ |
| | (0.00855) | (0.00453) | (0.00453) | (0.00504) | (0.00474) |
| $-\hat{MP}_{t-1}$ | $10.12^{**}$ | $8.879^{*}$ | $8.879^{*}$ | $8.136^{*}$ | $9.805^{**}$ |
| | (4.694) | (4.603) | (4.603) | (4.060) | (4.092) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $-0.00996$ | $-0.00994$ | $-0.00994$ | $-0.0123$ | $-0.00574$ |
| | (0.0208) | (0.0195) | (0.0195) | (0.0192) | (0.0198) |
| $R^2$ | 0.143 | 0.148 | 0.148 | 0.143 | 0.147 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,468,423 | 2,362,540 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.187^{***}$ | $-0.106^{***}$ | $-0.106^{***}$ | $-0.0857^{***}$ | $-0.0809^{***}$ |
| | (0.0169) | (0.00883) | (0.00883) | (0.00855) | (0.00877) |
| $-\hat{MP}_{t-1}$ | 6.762 | 4.060 | 4.060 | 3.751 | 5.077 |
| | (8.662) | (7.742) | (7.742) | (6.381) | (5.949) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $-0.00719$ | $-0.00757$ | $-0.00757$ | $-0.00623$ | 0.00192 |
| | (0.0226) | (0.0199) | (0.0199) | (0.0196) | (0.0201) |
| $R^2$ | 0.187 | 0.198 | 0.198 | 0.192 | 0.198 |
| Observations | 2,470,015 | 2,470,015 | 2,470,015 | 2,138,690 | 2,038,840 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level real business revenue to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of real business revenue, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 6. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A6: Response of Employment by Short-Term Loan Ratio, Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0649^{***}$ | $-0.0365^{***}$ | $-0.0365^{***}$ | $-0.0264^{***}$ | $-0.0317^{***}$ |
| | (0.00628) | (0.00449) | (0.00449) | (0.00482) | (0.00302) |
| $-\hat{MP}_{t-1}$ | $3.753^{**}$ | $3.135^{**}$ | $3.135^{**}$ | $2.796^{*}$ | $2.834^{**}$ |
| | (1.521) | (1.375) | (1.375) | (1.531) | (1.238) |
| Exposure $\times -\hat{MP}_{t-1}$ | −0.000298 | 0.000584 | 0.000584 | 0.00145 | 0.00263 |
| | (0.0165) | (0.0162) | (0.0162) | (0.0160) | (0.0164) |
| $R^2$ | 0.143 | 0.145 | 0.145 | 0.138 | 0.142 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,406,130 | 2,301,308 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.131^{***}$ | $-0.0805^{***}$ | $-0.0805^{***}$ | $-0.0639^{***}$ | $-0.0692^{***}$ |
| | (0.0104) | (0.00595) | (0.00595) | (0.00620) | (0.00519) |
| $-\hat{MP}_{t-1}$ | 2.915 | 1.664 | 1.664 | 0.723 | 0.188 |
| | (2.669) | (2.178) | (2.178) | (1.865) | (1.634) |
| Exposure $\times -\hat{MP}_{t-1}$ | 0.00231 | 0.00342 | 0.00342 | 0.00166 | 0.00390 |
| | (0.0218) | (0.0214) | (0.0214) | (0.0216) | (0.0215) |
| $R^2$ | 0.192 | 0.197 | 0.197 | 0.194 | 0.201 |
| Observations | 2,464,284 | 2,464,284 | 2,464,284 | 2,086,115 | 1,987,350 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level employment to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of the number of employees aged 26–65 between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times -\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as employment growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 7. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A7: Response of Average Wage by Short-Term Loan Ratio, Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0244^{***}$ | $0.00591^{*}$ | $0.00591^{*}$ | $0.0123^{***}$ | 0.00668 |
| | (0.00436) | (0.00333) | (0.00334) | (0.00395) | (0.00405) |
| $-\hat{MP}_{t-1}$ | $3.319^{**}$ | $2.521^{**}$ | $2.521^{**}$ | $3.372^{**}$ | $3.760^{***}$ |
| | (1.347) | (1.180) | (1.180) | (1.327) | (0.865) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $-0.0107$ | $-0.0103$ | $-0.0103$ | $-0.0105$ | $-0.0120$ |
| | (0.0113) | (0.0110) | (0.0110) | (0.0107) | (0.0113) |
| $R^2$ | 0.135 | 0.138 | 0.138 | 0.127 | 0.132 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,413,862 | 2,308,155 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.0319^{***}$ | $0.00925^{*}$ | $0.00925^{*}$ | $0.0171^{***}$ | $0.00929^{*}$ |
| | (0.00800) | (0.00499) | (0.00499) | (0.00509) | (0.00523) |
| $-\hat{MP}_{t-1}$ | $4.182^{**}$ | $3.063^{*}$ | $3.063^{*}$ | $3.573^{*}$ | $4.263^{**}$ |
| | (1.942) | (1.612) | (1.612) | (1.826) | (1.479) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $-0.00809$ | $-0.00775$ | $-0.00775$ | $-0.0117$ | $-0.0135$ |
| | (0.0159) | (0.0152) | (0.0152) | (0.0163) | (0.0172) |
| $R^2$ | 0.168 | 0.172 | 0.172 | 0.168 | 0.175 |
| Observations | 2,470,595 | 2,470,595 | 2,470,595 | 2,097,071 | 1,997,338 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average wage to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of the real average wage per employee, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 8. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A8: Response of Average Dividend by Short-Term Loan Ratio, Alternative Shock

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.138^{***}$ | 0.0153 | 0.0153 | $0.0588^{***}$ | $0.0578^{***}$ |
| | (0.0173) | (0.0110) | (0.0110) | (0.00797) | (0.00887) |
| $-\hat{MP}_{t-1}$ | 4.969 | 2.145 | 2.144 | 5.229 | 7.735 |
| | (7.173) | (6.876) | (6.876) | (6.133) | (4.999) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $0.0545^{*}$ | $0.0592^{**}$ | $0.0592^{**}$ | $0.0533^{*}$ | $0.0565^{**}$ |
| | (0.0279) | (0.0244) | (0.0244) | (0.0255) | (0.0265) |
| $R^2$ | 0.126 | 0.156 | 0.156 | 0.117 | 0.121 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,168,282 | 2,075,172 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.200^{***}$ | 0.0132 | 0.0132 | $0.0731^{***}$ | $0.0763^{***}$ |
| | (0.0293) | (0.0148) | (0.0148) | (0.0118) | (0.0132) |
| $-\hat{MP}_{t-1}$ | $13.95^{*}$ | 9.509 | 9.509 | $13.95^{*}$ | $19.87^{***}$ |
| | (7.605) | (6.891) | (6.891) | (7.112) | (5.141) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.00273 | 0.00557 | 0.00557 | 0.00399 | 0.00929 |
| | (0.0588) | (0.0557) | (0.0557) | (0.0569) | (0.0576) |
| $R^2$ | 0.152 | 0.181 | 0.181 | 0.158 | 0.165 |
| Observations | 2,470,337 | 2,470,337 | 2,470,337 | 1,893,886 | 1,805,476 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average dividend to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using an alternative measure of the monetary policy shock. The dependent variable is the cumulative growth rate of the real average dividend per individual shareholder, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the alternative expansionary monetary policy shock realised at $t-1$, constructed following Bauer and Swanson (2023) as described in Appendix A, and normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 9. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A9: Response of CPI-Deflated Business Revenue to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 14.74** | 13.96** | 13.96** | 12.45*** | 14.13*** |
| | (5.732) | (6.523) | (6.523) | (4.285) | (4.209) |
| $R^2$ | 0.146 | 0.148 | 0.148 | 0.145 | 0.148 |
| Observations | 2,861,168 | 2,861,168 | 2,861,168 | 2,471,847 | 2,365,248 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 9.737 | 3.125 | 3.125 | 1.812 | 3.791 |
| | (9.518) | (9.800) | (9.800) | (7.647) | (7.220) |
| $R^2$ | 0.191 | 0.197 | 0.197 | 0.194 | 0.200 |
| Observations | 2,473,447 | 2,473,447 | 2,473,447 | 2,141,882 | 2,041,359 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of firm-level real business revenue to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using the consumer price index (CPI) instead of the GDP deflator to deflate nominal values. The dependent variable is the cumulative growth rate of CPI-deflated business revenue between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Sample sizes differ slightly from the baseline because observations with absolute growth rates above 500% are excluded based on CPI-deflated rather than GDP-deflated values. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 2. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A10: Response of CPI-Deflated Average Wage to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 2.724** | 0.819 | 0.819 | 1.082 | −0.167 |
| | (0.996) | (0.945) | (0.945) | (1.068) | (0.809) |
| $R^2$ | 0.135 | 0.138 | 0.138 | 0.127 | 0.131 |
| Observations | 2,861,168 | 2,861,168 | 2,861,168 | 2,416,811 | 2,310,436 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 2.289 | 0.451 | 0.451 | 0.263 | −1.258 |
| | (1.704) | (2.036) | (2.036) | (1.796) | (1.245) |
| $R^2$ | 0.170 | 0.172 | 0.172 | 0.168 | 0.174 |
| Observations | 2,474,029 | 2,474,029 | 2,474,029 | 2,099,821 | 1,999,464 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of the firm-level average wage to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using the consumer price index (CPI) instead of the GDP deflator to deflate nominal values. The dependent variable is the cumulative growth rate of the CPI-deflated average wage per employee between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Sample sizes differ slightly from the baseline because observations with absolute growth rates above 500% are excluded based on CPI-deflated rather than GDP-deflated values. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 4. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A11: Response of CPI-Deflated Average Dividend to Monetary Policy Shocks

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| $-\hat{MP}_{t-1}$ | 14.95 | 6.777 | 6.777 | 10.13 | 10.10* |
| | (8.921) | (10.58) | (10.58) | (6.776) | (5.101) |
| $R^2$ | 0.129 | 0.156 | 0.156 | 0.118 | 0.122 |
| Observations | 2,861,168 | 2,861,168 | 2,861,168 | 2,170,361 | 2,076,762 |
| **Panel B:** $h = 1$ | | | | | |
| $-\hat{MP}_{t-1}$ | 30.12*** | 19.65** | 19.65** | 22.80*** | 25.64*** |
| | (6.453) | (8.238) | (8.238) | (6.257) | (4.924) |
| $R^2$ | 0.158 | 0.181 | 0.181 | 0.158 | 0.163 |
| Observations | 2,473,768 | 2,473,768 | 2,473,768 | 1,895,825 | 1,806,975 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports the cumulative response of the firm-level average dividend to an expansionary monetary policy shock, estimated from the local projection in Equation (2) using the consumer price index (CPI) instead of the GDP deflator to deflate nominal values. The dependent variable is the cumulative growth rate of the CPI-deflated average dividend per individual shareholder between $t-1$ and $t+h$. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points; its coefficient is the cumulative response in percent. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Sample sizes differ slightly from the baseline because observations with absolute growth rates above 500% are excluded based on CPI-deflated rather than GDP-deflated values. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 5. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A12: Response of CPI-Deflated Business Revenue by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0716^{***}$ | $-0.0461^{***}$ | $-0.0461^{***}$ | $-0.0368^{***}$ | $-0.0345^{***}$ |
| | (0.00779) | (0.00513) | (0.00513) | (0.00562) | (0.00518) |
| $-\hat{MP}_{t-1}$ | $14.52^{**}$ | $13.76^{**}$ | $13.76^{**}$ | $12.33^{***}$ | $14.02^{***}$ |
| | (5.714) | (6.506) | (6.506) | (4.234) | (4.145) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0147 | 0.0217 | 0.0218 | 0.0127 | 0.0118 |
| | (0.0166) | (0.0157) | (0.0157) | (0.0153) | (0.0160) |
| $R^2$ | 0.146 | 0.148 | 0.148 | 0.145 | 0.149 |
| Observations | 2,861,168 | 2,861,168 | 2,861,168 | 2,471,847 | 2,365,248 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.175^{***}$ | $-0.112^{***}$ | $-0.112^{***}$ | $-0.0910^{***}$ | $-0.0818^{***}$ |
| | (0.0162) | (0.00916) | (0.00916) | (0.00866) | (0.00784) |
| $-\hat{MP}_{t-1}$ | 9.389 | 2.950 | 2.950 | 1.735 | 3.719 |
| | (9.401) | (9.668) | (9.668) | (7.495) | (7.050) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0210 | 0.0198 | 0.0198 | 0.00816 | 0.00614 |
| | (0.0242) | (0.0238) | (0.0238) | (0.0243) | (0.0250) |
| $R^2$ | 0.191 | 0.198 | 0.198 | 0.194 | 0.200 |
| Observations | 2,473,447 | 2,473,447 | 2,473,447 | 2,141,882 | 2,041,359 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level real business revenue to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using the consumer price index (CPI) instead of the GDP deflator to deflate nominal values. The dependent variable is the cumulative growth rate of CPI-deflated business revenue between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Sample sizes differ slightly from the baseline because observations with absolute growth rates above 500% are excluded based on CPI-deflated rather than GDP-deflated values. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 6. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A13: Response of CPI-Deflated Average Wage by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | −0.0213*** | 0.00299 | 0.00299 | 0.00859** | 0.00464 |
| | (0.00467) | (0.00305) | (0.00305) | (0.00329) | (0.00274) |
| $-\hat{MP}_{t-1}$ | 2.849** | 0.937 | 0.937 | 1.209 | −0.0365 |
| | (1.025) | (0.978) | (0.978) | (1.123) | (0.837) |
| Exposure × $-\hat{MP}_{t-1}$ | −0.0164 | −0.0133 | −0.0133 | −0.0138 | −0.0140 |
| | (0.0119) | (0.0111) | (0.0111) | (0.0111) | (0.0113) |
| $R^2$ | 0.136 | 0.138 | 0.138 | 0.127 | 0.131 |
| Observations | 2,861,168 | 2,861,168 | 2,861,168 | 2,416,811 | 2,310,436 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | −0.0283*** | 0.00163 | 0.00163 | 0.00971* | 0.00681 |
| | (0.00913) | (0.00515) | (0.00515) | (0.00497) | (0.00453) |
| $-\hat{MP}_{t-1}$ | 2.326 | 0.479 | 0.479 | 0.341 | −1.182 |
| | (1.860) | (2.165) | (2.165) | (1.851) | (1.275) |
| Exposure × $-\hat{MP}_{t-1}$ | −0.00707 | −0.00315 | −0.00315 | −0.00839 | −0.00810 |
| | (0.0239) | (0.0221) | (0.0221) | (0.0222) | (0.0220) |
| $R^2$ | 0.170 | 0.172 | 0.172 | 0.168 | 0.174 |
| Observations | 2,474,029 | 2,474,029 | 2,474,029 | 2,099,821 | 1,999,464 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average wage to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using the consumer price index (CPI) instead of the GDP deflator to deflate nominal values. The dependent variable is the cumulative growth rate of the CPI-deflated average wage per employee between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure × $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Sample sizes differ slightly from the baseline because observations with absolute growth rates above 500% are excluded based on CPI-deflated rather than GDP-deflated values. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 8. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A14: Response of CPI-Deflated Average Dividend by Short-Term Loan Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.125^{***}$ | 0.0104 | 0.0104 | $0.0541^{***}$ | $0.0567^{***}$ |
| | (0.0180) | (0.0122) | (0.0122) | (0.00948) | (0.0108) |
| $-\hat{MP}_{t-1}$ | 14.31 | 6.044 | 6.044 | 9.321 | $9.260^{*}$ |
| | (8.951) | (10.53) | (10.53) | (6.684) | (4.996) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $0.0543^{*}$ | $0.0833^{***}$ | $0.0833^{***}$ | $0.0875^{***}$ | $0.0922^{***}$ |
| | (0.0285) | (0.0211) | (0.0211) | (0.0223) | (0.0218) |
| $R^2$ | 0.129 | 0.156 | 0.156 | 0.118 | 0.122 |
| Observations | 2,861,168 | 2,861,168 | 2,861,168 | 2,170,361 | 2,076,762 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.177^{***}$ | 0.00223 | 0.00223 | $0.0617^{***}$ | $0.0675^{***}$ |
| | (0.0279) | (0.0166) | (0.0166) | (0.0129) | (0.0135) |
| $-\hat{MP}_{t-1}$ | $29.88^{***}$ | $19.34^{**}$ | $19.34^{**}$ | $22.43^{***}$ | $25.26^{***}$ |
| | (6.324) | (8.131) | (8.131) | (6.026) | (4.598) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.00866 | 0.0346 | 0.0346 | 0.0396 | 0.0422 |
| | (0.0740) | (0.0645) | (0.0645) | (0.0647) | (0.0667) |
| $R^2$ | 0.159 | 0.181 | 0.181 | 0.158 | 0.163 |
| Observations | 2,473,768 | 2,473,768 | 2,473,768 | 1,895,825 | 1,806,975 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average dividend to an expansionary monetary policy shock varies with firms' pre-existing short-term loan ratio, estimated from Equation (3) using the consumer price index (CPI) instead of the GDP deflator to deflate nominal values. The dependent variable is the cumulative growth rate of the CPI-deflated average dividend per individual shareholder between $t-1$ and $t+h$. Exposure is the short-term loan ratio, defined as short-term loans divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Sample sizes differ slightly from the baseline because observations with absolute growth rates above 500% are excluded based on CPI-deflated rather than GDP-deflated values. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding baseline estimates are reported in Table 9. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A15: Response of Business Revenue by Leverage Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.223^{***}$ | $-0.210^{***}$ | $-0.209^{***}$ | $-0.198^{***}$ | $-0.200^{***}$ |
| | (0.00679) | (0.00733) | (0.00733) | (0.00824) | (0.00832) |
| $-\hat{MP}_{t-1}$ | $17.25^{***}$ | $16.34^{**}$ | $16.34^{**}$ | $15.43^{***}$ | $17.11^{***}$ |
| | (5.116) | (6.080) | (6.080) | (4.217) | (4.132) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0190 | 0.0162 | 0.0162 | 0.0123 | 0.0113 |
| | (0.0152) | (0.0145) | (0.0145) | (0.0152) | (0.0154) |
| $R^2$ | 0.153 | 0.154 | 0.154 | 0.150 | 0.153 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,468,423 | 2,362,540 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.403^{***}$ | $-0.375^{***}$ | $-0.375^{***}$ | $-0.352^{***}$ | $-0.349^{***}$ |
| | (0.0162) | (0.0149) | (0.0149) | (0.0150) | (0.0155) |
| $-\hat{MP}_{t-1}$ | 15.97 | 8.559 | 8.559 | 7.659 | 9.578 |
| | (9.529) | (9.673) | (9.673) | (7.173) | (7.363) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0120 | 0.00113 | 0.00114 | $-0.00422$ | $-0.00405$ |
| | (0.0245) | (0.0230) | (0.0230) | (0.0261) | (0.0270) |
| $R^2$ | 0.196 | 0.203 | 0.203 | 0.199 | 0.204 |
| Observations | 2,470,015 | 2,470,015 | 2,470,015 | 2,138,690 | 2,038,840 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level real business revenue to an expansionary monetary policy shock varies with firms' pre-existing leverage ratio, estimated from Equation (3) using the leverage ratio instead of the short-term loan ratio as the exposure measure. The dependent variable is the cumulative growth rate of real business revenue, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the leverage ratio, defined as total liabilities divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as revenue growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding estimates using the short-term loan ratio are reported in Table 6. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A16: Response of Employment by Leverage Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0847^{***}$ | $-0.0717^{***}$ | $-0.0717^{***}$ | $-0.0654^{***}$ | $-0.0702^{***}$ |
| | (0.00491) | (0.00461) | (0.00461) | (0.00382) | (0.00355) |
| $-\hat{MP}_{t-1}$ | $5.059^{***}$ | $3.704^{**}$ | $3.704^{**}$ | $3.564^{**}$ | 2.018 |
| | (1.354) | (1.640) | (1.640) | (1.591) | (1.297) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0253 | 0.0224 | 0.0224 | 0.0182 | 0.0184 |
| | (0.0179) | (0.0179) | (0.0179) | (0.0175) | (0.0173) |
| $R^2$ | 0.144 | 0.146 | 0.146 | 0.138 | 0.143 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,406,130 | 2,301,308 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.183^{***}$ | $-0.161^{***}$ | $-0.161^{***}$ | $-0.146^{***}$ | $-0.148^{***}$ |
| | (0.00948) | (0.00863) | (0.00863) | (0.00792) | (0.00745) |
| $-\hat{MP}_{t-1}$ | 5.194 | 1.449 | 1.449 | 0.985 | $-1.284$ |
| | (3.197) | (3.728) | (3.728) | (2.840) | (1.835) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.0380 | 0.0315 | 0.0315 | 0.0207 | 0.0226 |
| | (0.0245) | (0.0235) | (0.0235) | (0.0249) | (0.0240) |
| $R^2$ | 0.194 | 0.198 | 0.198 | 0.196 | 0.203 |
| Observations | 2,464,284 | 2,464,284 | 2,464,284 | 2,086,115 | 1,987,350 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of firm-level employment to an expansionary monetary policy shock varies with firms' pre-existing leverage ratio, estimated from Equation (3) using the leverage ratio instead of the short-term loan ratio as the exposure measure. The dependent variable is the cumulative growth rate of the number of employees aged 26–65 between $t-1$ and $t+h$. Exposure is the leverage ratio, defined as total liabilities divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as employment growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding estimates using the short-term loan ratio are reported in Table 7. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A17: Response of Average Wage by Leverage Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0212^{***}$ | $-0.00721$ | $-0.00722$ | $-0.000635$ | $-0.00461$ |
| | (0.00502) | (0.00481) | (0.00481) | (0.00513) | (0.00517) |
| $-\hat{MP}_{t-1}$ | $6.383^{***}$ | $4.268^{***}$ | $4.268^{***}$ | $4.990^{***}$ | $3.778^{***}$ |
| | (1.178) | (1.172) | (1.172) | (1.424) | (1.085) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | $-0.0149$ | $-0.0180$ | $-0.0180$ | $-0.0173$ | $-0.0181$ |
| | (0.0118) | (0.0117) | (0.0117) | (0.0131) | (0.0138) |
| $R^2$ | 0.136 | 0.138 | 0.138 | 0.127 | 0.132 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,413,862 | 2,308,155 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.0405^{***}$ | $-0.0228^{***}$ | $-0.0228^{***}$ | $-0.00967$ | $-0.0143^{*}$ |
| | (0.00778) | (0.00706) | (0.00706) | (0.00693) | (0.00676) |
| $-\hat{MP}_{t-1}$ | $8.373^{***}$ | $5.452^{*}$ | $5.452^{*}$ | $5.774^{**}$ | $4.236^{*}$ |
| | (2.837) | (2.939) | (2.939) | (2.411) | (2.277) |
| Exposure $\times$ $-\hat{MP}_{t-1}$ | 0.00397 | 0.000483 | 0.000485 | $-0.00482$ | $-0.00524$ |
| | (0.0267) | (0.0258) | (0.0258) | (0.0262) | (0.0265) |
| $R^2$ | 0.169 | 0.172 | 0.172 | 0.169 | 0.175 |
| Observations | 2,470,595 | 2,470,595 | 2,470,595 | 2,097,071 | 1,997,338 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average wage to an expansionary monetary policy shock varies with firms' pre-existing leverage ratio, estimated from Equation (3) using the leverage ratio instead of the short-term loan ratio as the exposure measure. The dependent variable is the cumulative growth rate of the real average wage per employee, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the leverage ratio, defined as total liabilities divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times$ $-\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average wage growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding estimates using the short-term loan ratio are reported in Table 8. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table A18: Response of Average Dividend by Leverage Ratio

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| **Panel A:** $h = 0$ | | | | | |
| Exposure | $-0.0725^{***}$ | 0.0116 | 0.0116 | $0.0380^{***}$ | $0.0395^{**}$ |
| | (0.0125) | (0.0119) | (0.0119) | (0.0114) | (0.0135) |
| $-\hat{MP}_{t-1}$ | $16.13^{*}$ | 8.058 | 8.058 | $11.11^{*}$ | $10.99^{**}$ |
| | (9.028) | (10.64) | (10.64) | (5.933) | (4.371) |
| Exposure $\times -\hat{MP}_{t-1}$ | $0.0526^{**}$ | $0.0439^{*}$ | $0.0439^{*}$ | $0.0700^{**}$ | $0.0734^{**}$ |
| | (0.0228) | (0.0226) | (0.0226) | (0.0250) | (0.0253) |
| $R^2$ | 0.130 | 0.157 | 0.157 | 0.121 | 0.125 |
| Observations | 2,857,459 | 2,857,459 | 2,857,459 | 2,168,282 | 2,075,172 |
| **Panel B:** $h = 1$ | | | | | |
| Exposure | $-0.255^{***}$ | $-0.149^{***}$ | $-0.149^{***}$ | $-0.0728^{***}$ | $-0.0703^{**}$ |
| | (0.0295) | (0.0272) | (0.0272) | (0.0245) | (0.0254) |
| $-\hat{MP}_{t-1}$ | $34.76^{***}$ | $23.44^{**}$ | $23.44^{**}$ | $26.25^{***}$ | $28.97^{***}$ |
| | (6.323) | (8.115) | (8.115) | (5.635) | (4.096) |
| Exposure $\times -\hat{MP}_{t-1}$ | 0.0584 | 0.0476 | 0.0476 | 0.0715 | 0.0735 |
| | (0.0673) | (0.0629) | (0.0629) | (0.0632) | (0.0647) |
| $R^2$ | 0.161 | 0.184 | 0.184 | 0.162 | 0.167 |
| Observations | 2,470,337 | 2,470,337 | 2,470,337 | 1,893,886 | 1,805,476 |
| *Model Specifications* | | | | | |
| Firm FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Age FE | | ✓ | ✓ | ✓ | ✓ |
| Profit Margin | | | ✓ | ✓ | ✓ |
| Dynamic Controls | | | | ✓ | ✓ |
| Extended Policy Lags | | | | | ✓ |

*Notes:* This table reports how the cumulative response of the firm-level average dividend to an expansionary monetary policy shock varies with firms' pre-existing leverage ratio, estimated from Equation (3) using the leverage ratio instead of the short-term loan ratio as the exposure measure. The dependent variable is the cumulative growth rate of the real average dividend per individual shareholder, deflated by the GDP deflator, between $t-1$ and $t+h$. Exposure is the leverage ratio, defined as total liabilities divided by total assets, in percent, measured at $t-2$ so that it is determined prior to the realisation of the shock. $-\hat{MP}_{t-1}$ denotes the expansionary monetary policy shock realised at $t-1$, normalised to lower the overnight interbank rate by 12.5 basis points. The coefficient on $-\hat{MP}_{t-1}$ is the cumulative response in percent for a firm with zero exposure. The coefficient on Exposure $\times -\hat{MP}_{t-1}$ is the additional cumulative response, in percentage points, associated with a one-percentage-point higher exposure. Panel A reports estimates at $h = 0$ and Panel B at $h = 1$. Column (1) includes only firm fixed effects. Column (2) adds firm age fixed effects. Column (3) further controls for the firm's net profit margin lagged two periods. Column (4) adds dynamic controls: two lags of the dependent variable, measured as average dividend growth from $t-2$ to $t-1$ and from $t-3$ to $t-2$, and the monetary policy shock at $t-2$. Column (5) additionally includes the monetary policy shocks at $t-3$ and $t-4$. The number of observations declines in Columns (4) and (5) because the additional lags require longer firm histories. Standard errors, two-way clustered at the firm and year levels, are reported in parentheses. The corresponding estimates using the short-term loan ratio are reported in Table 9. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.